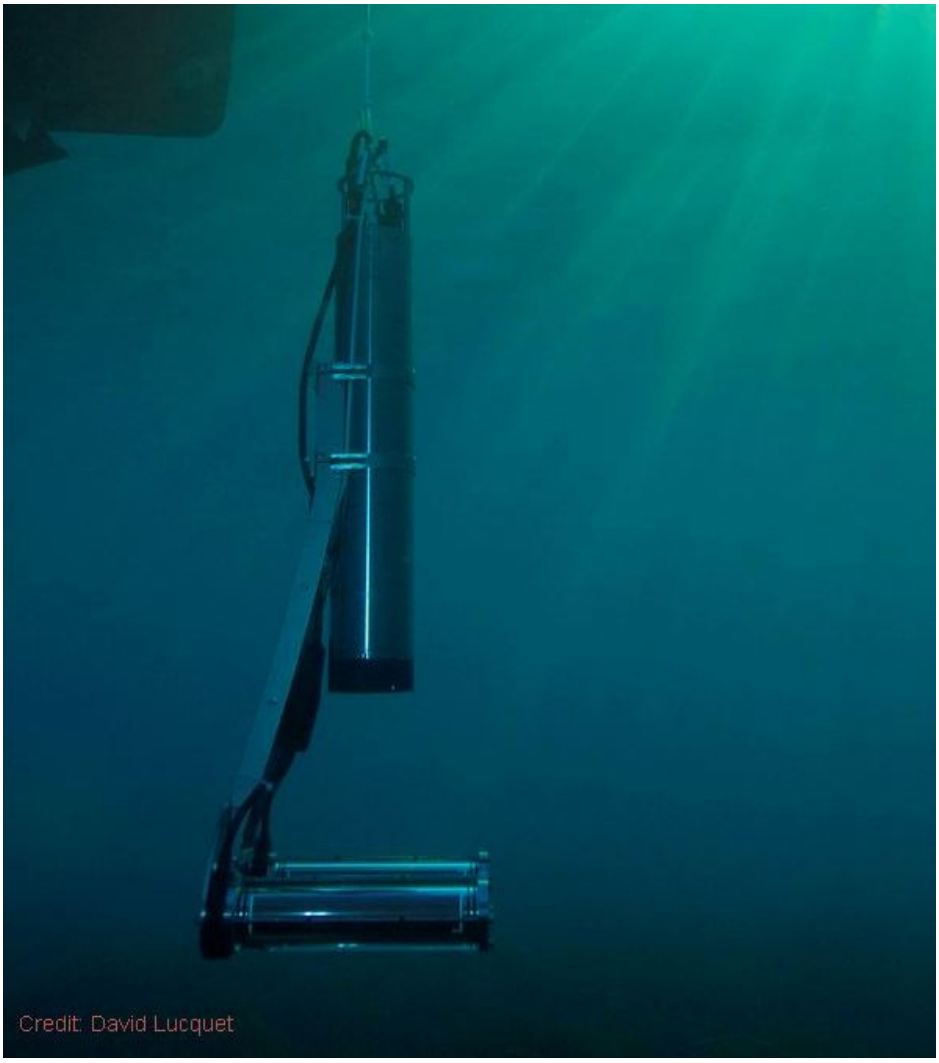
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UVP5 USER MANUAL

(standard and HD)

v. 2021/02/01



With the assistance of
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WARNING!

If you suspect that the UVP5 main pressure case or any of the light units has flooded, use EXTREME CAUTION around the instrument. The UVP5 can operate at depths of up to 6000 meters but the maximum depth is LIMITED by the PRESSURE SENSOR installed on your instrument (check provided documentation and notice on the main pressure case).

If a light leaked at depth it might remain highly pressurized when recovered and cause the end cap to be launched from the pressure case with extreme force if retaining screws are removed.

The main pressure case is equipped with a relief valve that should prevent any over pressure.

An indication for flooding is that the instrument stops operating or that there is a short-circuit condition in the instrument. An instrument flooded with salt water will short all of the connector pins together. An electric continuity test between random pins on any of the bulkheads may confirm this suspicion.

In case you suspect any flooding, place the instrument in a safe location and contact HYDROPTIC for further instructions.

Never stay in front of the end caps!

CONTINUE AT YOUR ON RISK!

Even if the relief valve should prevent over pressure in the main case, you should try to depressurize the main pressure case slowly backing off one of the valve located in the center of the connector end cap. The valve only has to be loosened so that the face seal O ring is no longer sealed against the end cap. When all of the pressure has been released the instrument can be stored and safely shipped to Hydroptic for repairs.

You can also open the pressure case by the connector end, disconnect the battery and rinse with fresh water under Hydroptic supervision.

There is nothing you can do if you suspect light flooded. Just prevent over accident.

Instruments

Do not leave instruments in direct sunlight. Direct sunlight can easily increase the internal temperature of the instrument beyond its maximum rating.

LIGHT UNITS

Even if rated and tested for 6000m use, the glass cylinders of the light unit are fragile. Extreme care must be applied to prevent any scratch or shock on these units. If you suspect any damage, stop using UVP and contact Hydroptic.

CONNECTIONS

Handle electrical terminations carefully, as they are not designed to withstand strain. Disconnect the cables from the components by pulling on the connector heads and not the cables. Do not twist the connector while pulling, as this will damage the connector pins.

Do not use petroleum-based lubricants on Subconn® connectors. Connectors should be free of dirt and lightly lubricated before mating.

Hydroptic recommends using the provided silicone grease on the male pins prior to connection.

TROUBLESHOTING

While checking voltages with a multimeter, use extreme care to avoid shorting the probe leads. A shorted power supply or battery can output many amperes of current, potentially harming the user, starting fires, or damaging equipment.

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1 Introduction

The manual enables the user to be directed pertaining to the contents of the crates received from Hydroptic and that can contain variable parts and optional spare parts. The user will be guided through the unpacking and system setup and fine-tuning prior to the system being sea bound. In the same time basic functions and features are explained for a quick understanding and ownership of the UVP. More manuals are available in the PIQv website.

2 UVP5 Standard and UVP5HD High Definition

UVP5 s/n above 200 are High Definition systems. The camera is 4 Mp and the acquisition frequency is improved to up to 20 Hz. HD systems have longer battery autonomy than the standard system.

The operations are identical for both versions and this manual is adapted for both.

3 Regulations (customs, safety)

3.1 HS code

3.1.1 International

901580

3.1.2 French

9015801900

3.2 US custom ECCN

The UVP5 US 8A992.

3.3 Battery UN

UVP5 Li-Ion batteries are located inside the main pressure case. Their UN number which must be indicated for transport is thus **3481: "Lithium Ion Batteries contained in equipment"**.

Weight: 1.085 Kg
Power: 29.4 Vdc / 6.8 A

The official documents which must be provided to the transporter are in annex.

4 UVP assembly procedure

4.1 Packing list

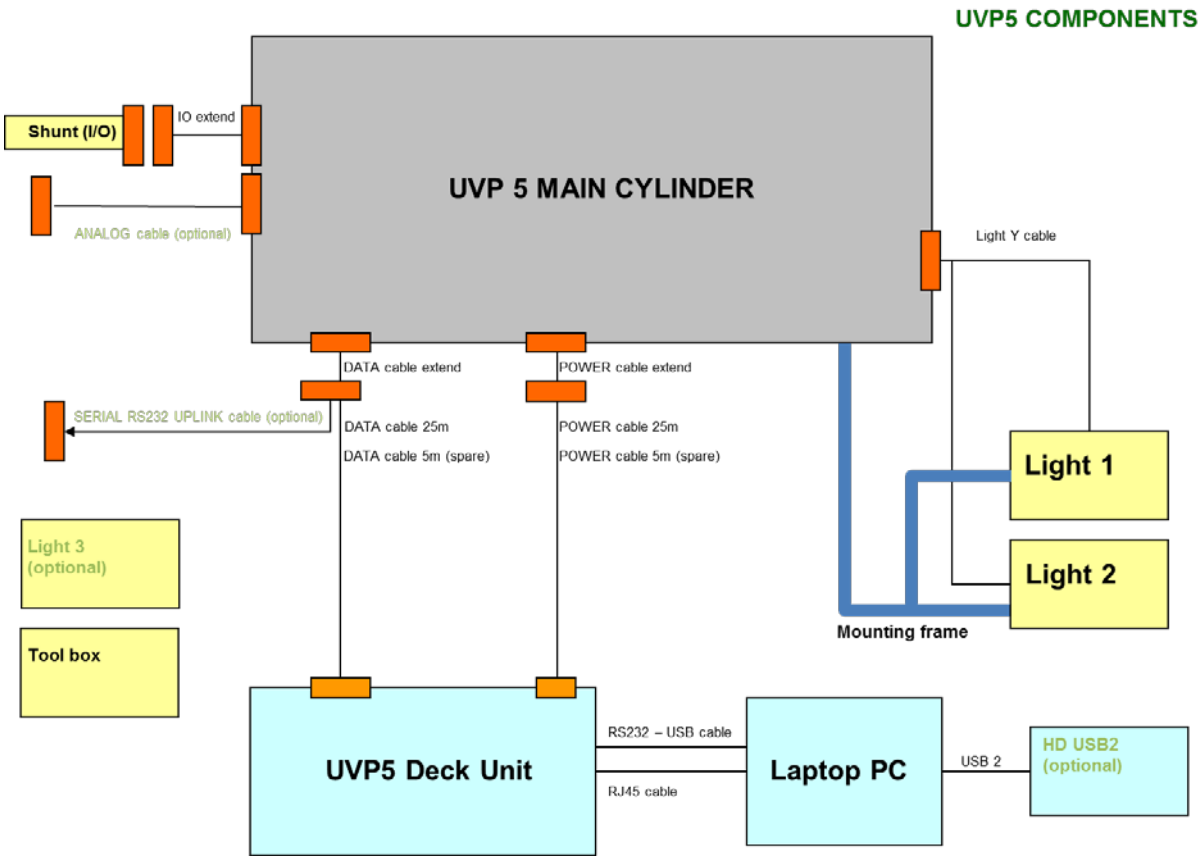
The instruments are provided with packing lists. It is very important to unpack the transport case checking the presence of all items.

4.2 Description of components

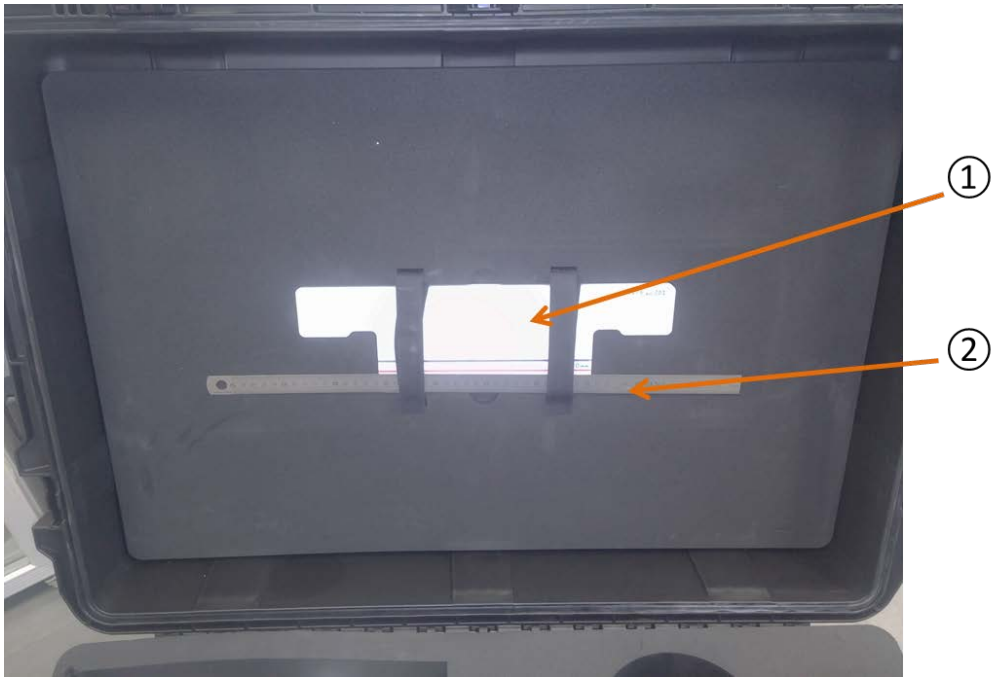
This section of the manual describes the detailed contents of the shipping crates. The location of the parts will allow the system setup to be easily performed through the identification of the different parts shown in the section.

The single crate includes three foam protected levels in order to have all the contents in the same container. Each level can be removed individually in order to access the different parts that are placed on each level. The levels are shown hereunder.

The diagram below presents the different components of the UVP5.



4.2.1 UVP5HD Crate Lid



① T shape Light alignment JIG
② 50 cm Ruler

4.2.2 UVP5HD Crate Level 1

The first level of the crate corresponds to the elements that can be seen upon the opening of the crate. The first level includes the following components:

- ① Two frame mounted LED lighting units
- ② Power cable, USB cable and RJ45 cable
- USB key with complete set of Documentation and software applications (appendix) / Equipment manuals and calibration report
- Power and IO shunts
- ③ Deck Unit
- ④ CTD Mounting kit (Optional)
- ⑤ Stations Notebook (Optional)
- ⑥ Molikote 44 Grease for connectors
- CALIBRATION REPORT



4.2.3 UVP5HD Crate Level 2

The second level of the crate corresponds to the elements that can be reached once the first level has been removed from the crate. The second level includes the following components:

- Sea cable 5 m and 25 m
- Power supply cable 5m and 25 m
- Power supply extension cable 1 m
- Sea Cable extension cable 1 m
- IO cable extension 1 m
- Light Cable

Note: Some parts of the third level can be seen from the second level of the crate. It is nevertheless recommended to access the third level parts once and only once the second level has been removed.



4.2.4 UVP5HD Crate Level 3

The third level of the crate corresponds to the elements that can be reached after the second level has been removed from the crate. The third level includes the following components:

- ① Pressure Case containing a smart camera, batteries, pressure sensor and its electronics.
- ② Pressure Case support piece
- ③ Pressure Case Attachment brackets
- ④ UVP5 installation kit including Tool box with screws, washers, and damper
- ⑤ Dedicated laptop compartment



4.2.5 Optional Spare Parts Crate Contents

The UVP5HD can be supplied with an additional set of parts (depending on the configuration and pre-defined sales agreement). The parts are placed in an additional crate and can include (but not limited to) the following components:

- SPARE Lighting Unit
- SPARE Electronic Components
- SPARE Ethernet Switch
- SPARE UVP5std or UVP5hd mother board
- SPARE Waterproof Connectors
- SPARE Flash Drive containing CAMERA OS
- OPTIONAL CTD rosette mounting KIT
- OPTIONAL CTD analog cable
- OPTIONAL CTD serial uplink cable
- OPTIONAL separate frame

4.3 System Setup

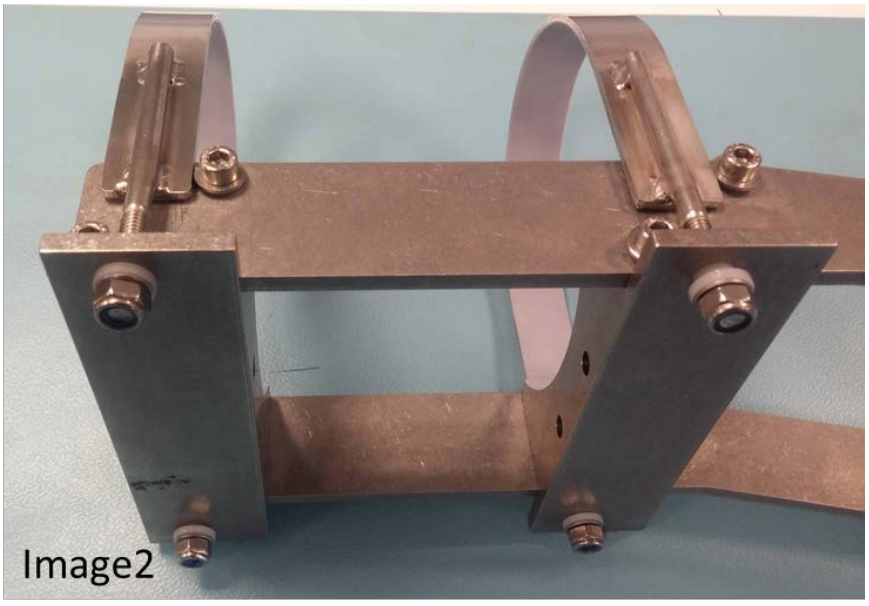
4.3.1 Assembling

In order to start the system setup, the parts need to be removed from the crate and placed on a safe and clean workplace. The installation kit needs to be available at all times. The setup sequence is detailed in the following step by step procedures.

Prepare the two camera attachment brackets①, the camera support piece② and 4 M6 plastic screw guides, 4 M6 washers, 4 M6 nuts. Image1



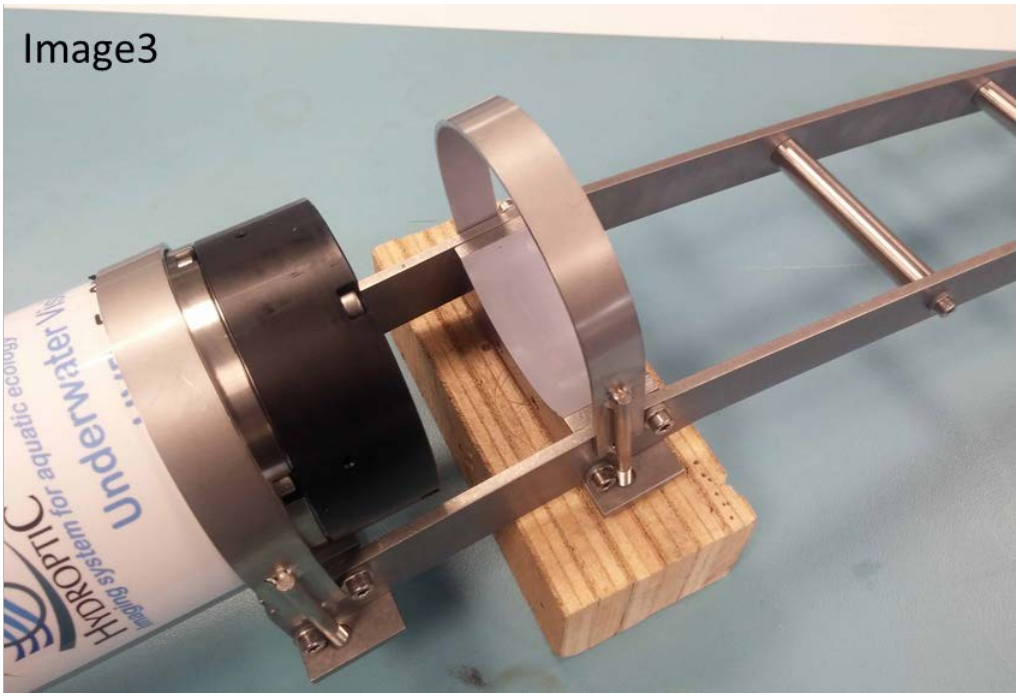
Insert both brackets, then the plastic guides, the washers and slightly tighten the nuts in order to leave enough mechanical play to slide the Camera unit through the brackets.Image2

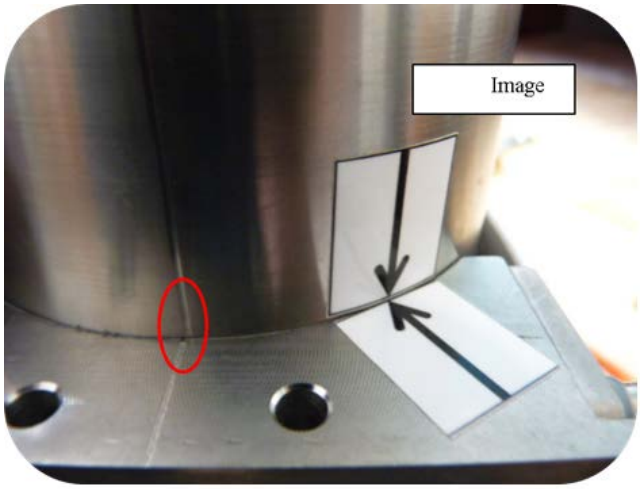


With the help of a buddy, slide the Camera Unit through the attachment brackets Image3 until both arrows match (Image4)

The distance between the Pressure case and the lighting units is pre-set at the factory. The distance is set at +/- 365 millimetres from the black porthole to the centre of the lighting units. THIS DISTANCE CAN VARY FROM UNIT TO UNIT. REFER TO CALIBRATION REPORT PROVIDED WITH THE INSTRUMENT. Two labelled arrows are used for the alignment and distance adjustment. (Image 4)

Two marks are used to align the tube with the lighting unit in the event the stickers peel off. Both marks must match on the tube and the support cradle.





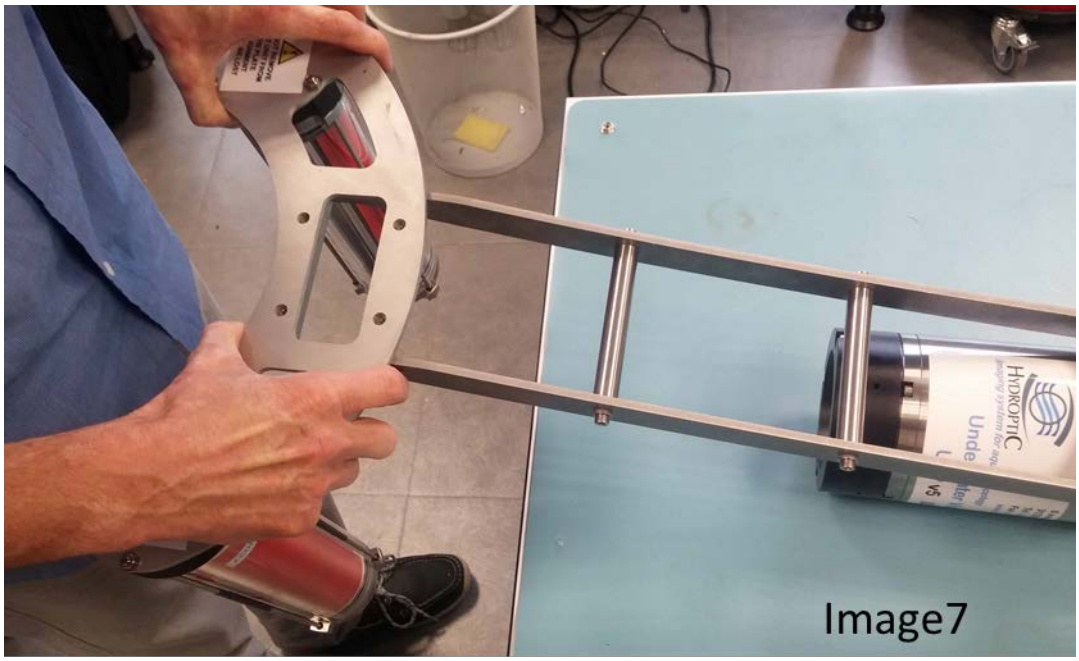
Tighten all four nuts to ensure the position of the camera unit.Image5

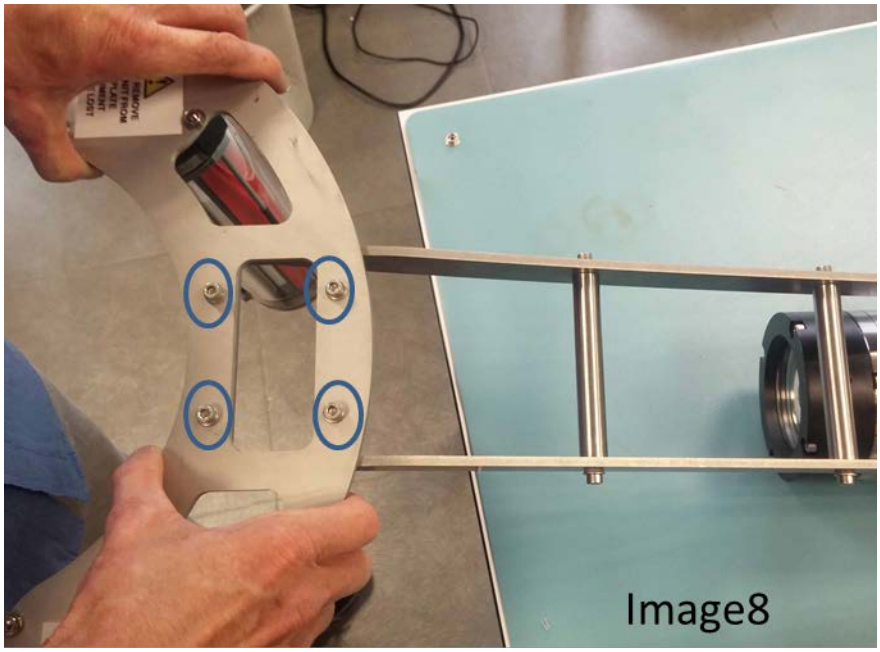


Turn the assembly upside-down image6



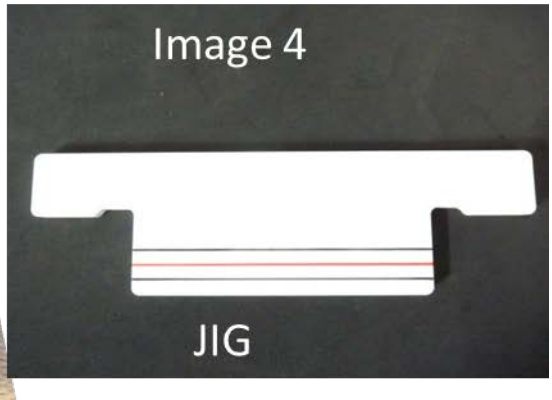
Place and hold the frame mounted lighting unit onto the arm Image 7 while the other buddy is inserting the plastic screw guides with the washers and finally tighten the screws Image 8





The distance can be checked and set up with the help of the alignment JIG by positioning it on the lighting unit as shown hereunder. The distance is indicated in the Qualification report provided with the instrument (ie 210mm for UVP5hd models).





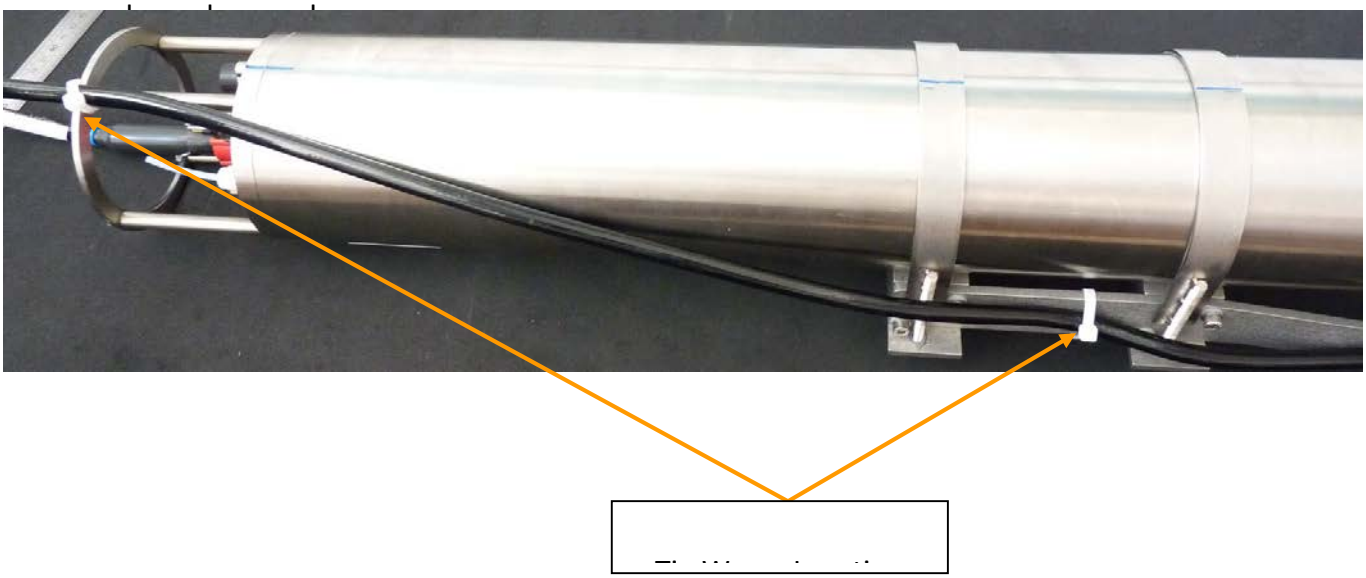
The mechanical assembly of the UVP5 is now complete. The cable connections need to be performed.



Connect the lighting unit cable on both lighting units.



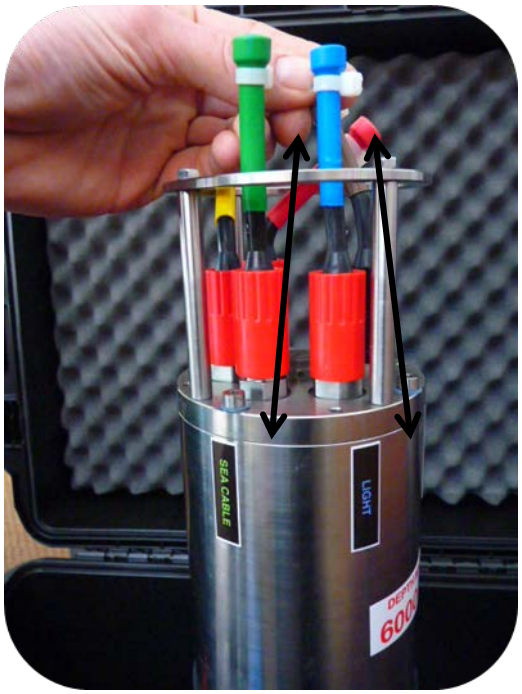
Place the tie-wraps onto the lighting unit cable along the Pressure Case support piece

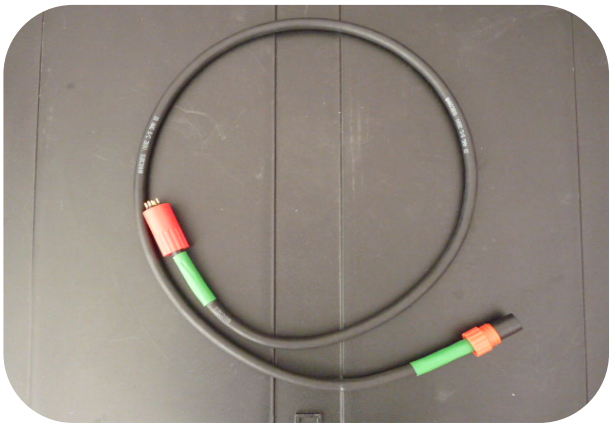




4.3.1.1.1 Cables and socket colour code

Each dummy, shunt and cable is colour marked and matches its connector.





4.3.1.1.2 Colour code

Ring Color	Socket	Cable in
Green	Subconn MCBH 8F	Data Cable
Blue	Subconn MCBH 4F	Light Cable
Red	Subconn MCBH 4M	Power Cable
Yellow	Subconn MCBH 6F	IO Cable
Brown	[illegible]3H 6H	CTD Cable

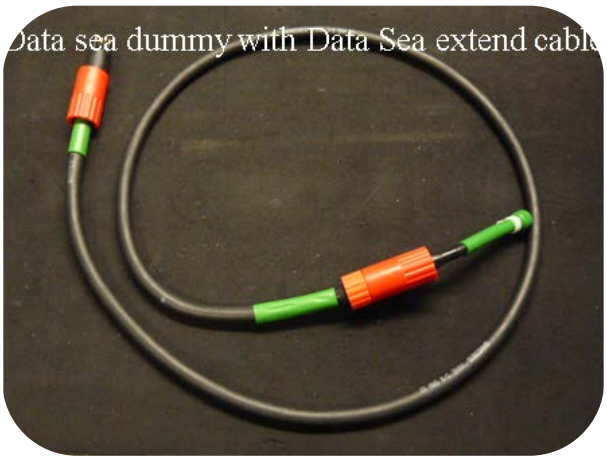
4.3.1.1.3 DISTINGUISHING DUMMY from SHUNT

SHUNT (active dummies) can be differentiated by their coloured ring of heat shrink (red for POWER, yellow for I/O) or their wider diameter (S/N above 8) added on the locking sleeve.

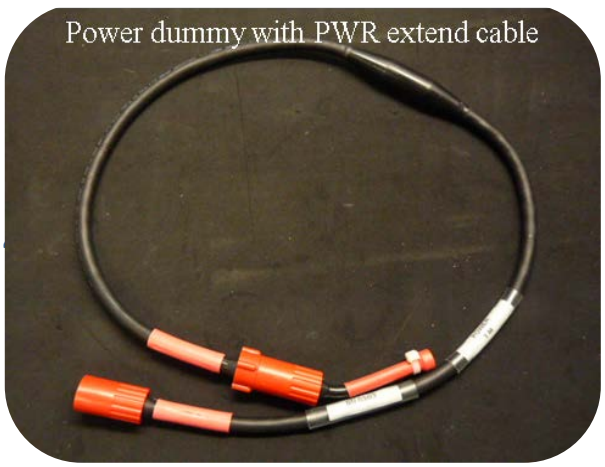
4.3.1.1.4 Recommendations

Titanium sockets

Shunt and dummy of Power, Sea cable and IO should be used in addition with their extend cable. It will prevent bad maneuver on the Titanium socket and lessen their worn out.







WARNING: DO NOT REMOVE LIGHT UNITS FROM THE CRESCENT PLATE OTHERWISE THE LED BEAM ALIGNMENT WILL BE LOST!



It is recommended to check the light alignment using the "T" JIG on a daily base when at sea.

5 Quick test to be happy when receiving the UVP5 from Hydroptic

This quick test permits user to check UVP5 is workable on reception from Hydroptic. We have programmed an I/O sequence (see below) to allow easy start of acquisition for this specific purpose. It also permits user to perform its first field profile using default settings preset in factory.

PLEASE FOLLOW CAREFULLY ALL STEPS and TIMINGS!

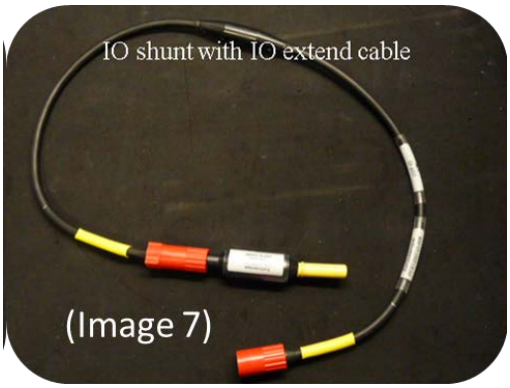
1. Check that LIGHT CABLE is connected.
2. Remove the power dummy (image5) from the top cap, plug it at the end of the power extension. Connect the power extension and replace the power dummy by the power shunt.





2 Remove the IO dummy from the top cap and install the IO extension. Plug in the IO shunt (image 7).





3 Acquisition starts 1 minute after plugging the IO shunt. Both light units are flashing after that time laps indicating that images are acquired and processed by the UVP5. Place your hand in front of the camera window. Flashing frequency will increase. => UVP is working fine.

4 Wait 2 minutes and remove the IO shunt from the extend cable to stop acquisition. Plug in the IO dummy.

5 Wait **more than 1minute** and replace the power shunt with the power dummy.

When the power shunt is removed, **ALL** power units in the pressure case are turned off instantaneously.

If the power shunt is removed directly after the IO shunt, it will eventually cause the camera OS to crash! **Respect the indicated timing of the operations.**

6 Connecting the UVP5hd to the Deck unit

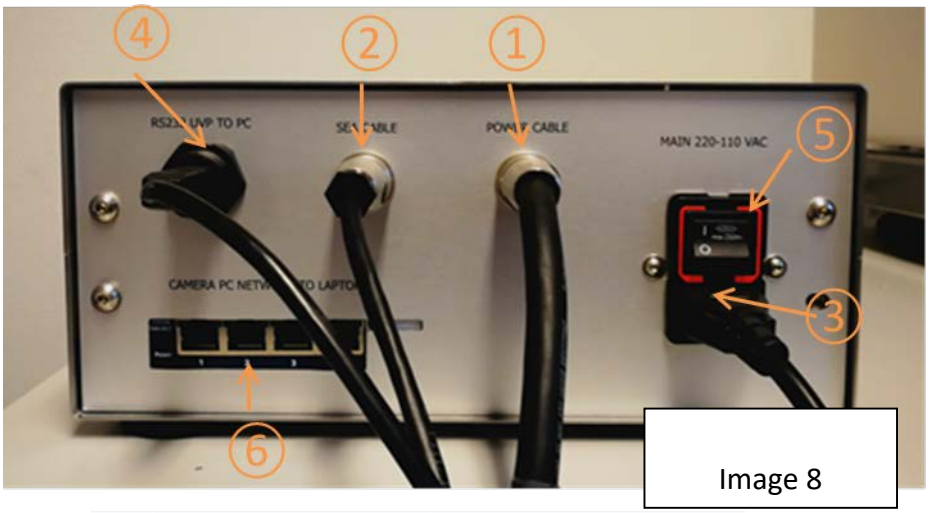


Image 8

No	Socket	Cable in
1	Amphenol 4 pts « Power Cable »	Power cable (from Pressure case)
2	Amphenol 8 pts « Sea Cable »	Data cable (from Pressure case)
3	« Main 220/110 VAC »	Main power plug
4	« RS232 user to PC »	RS232 cable (to PC)
6	« Ethernet to network PC »	RJ45 cable (to PC)

1) Plug the Power extend to the UVP5HD “Power” socket.
2) Plug in the 25 or the 5 meters power cable (red marked) to the extend
3) Plug the other end (amphenol) the UVP “Power Cable” socket of the Deck Unit ①

4) Plug the DATA extend to the UVP5HD “Data” socket.
5) Plug in the 25 or the 5 meters DATA cable (green marked) to the extend
6) Plug the other end (amphenol) the UVP “DATA Cable” socket of the Deck Unit ②

7) Plug the cable into the power outlet ③
8) Plug the USB cable ④ onto the laptop.
9) Plug the Ethernet cable between the laptop and the Deck Unit
10) Turn on the main power switch ⑤

7 Connecting the UVPstd to the Deck unit

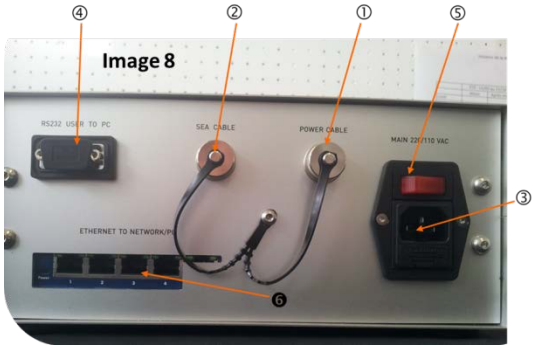


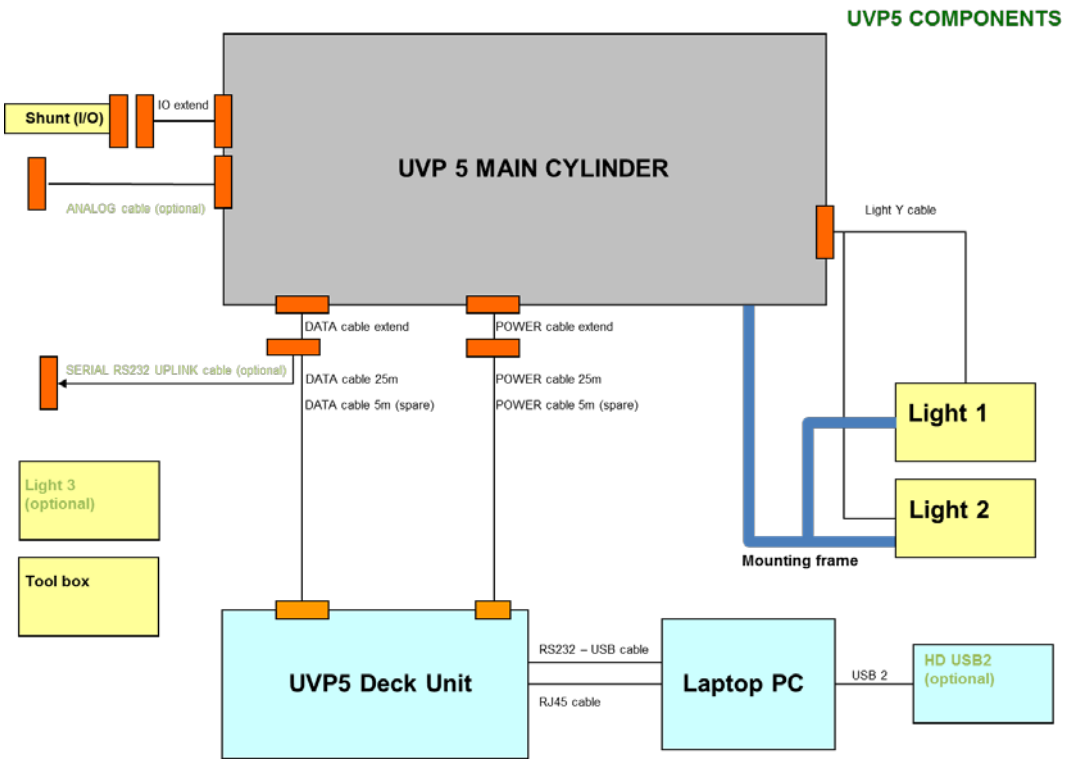
Image 8

No	Socket	Cable in
1	Amphenol 4 pts « Power Cable »	Power cable (from Pressure case)
2	Amphenol 8 pts « Sea Cable »	Data cable (from Pressure case)
3	« Main 220/110 VAC »	Main power plug
4	« RS232 user to PC »	RS232 cable (to PC)
6	« Ethernet to network PC »	RJ45 cable (to PC)

1) Plug the Power extend to the UVP5 “Power” socket.
2) Plug in the 25 or the 5-meter power cable (red marked) to the extend
3) Plug the other end (amphenol) the UVP “Power Cable” socket of the Deck Unit ①

4) Plug the DATA extend to the UVP5 “Data” socket.
5) Plug in the 25 or the 5 meters DATA cable (green marked) to the extend
6) Plug the other end (amphenol) the UVP “DATA Cable” socket of the Deck Unit ②

7) Plug the cable into the power outlet ③
8) Plug the Serial cable ④ with its USB adaptor onto the laptop.
9) Plug the Ethernet cable between the laptop and the Deck Unit
10) Turn on the main power switch (can be kept ON) ⑤



7.1 Computer network settings

The computer network configuration must be set to fixed IP (no DHCP).

- 193.49.112.130
- 255.255.255.0
- 193.49.112.1

7.2 UVP5 IP address

The UVP5 computer/camera default IP address is 193.49.112.100.

The camera can be networked only when powered ON.

8 Power

In order to secure the operation of the UVP5, the user should monitor the battery voltage to be sure that the remaining power (voltage) is sufficient for the duration of the planed image acquisitions.

8.1 Power sources

The UVP5 contains a Lithium-Ion 29.4/6.8A battery weighting 1085g. This battery is considered as a dangerous good and should be declared when shipping the UVP5HD (see appendix).

The UVP5 can thus be powered on its internal battery when performing image acquisition and process. It can also be powered by an external unit (additional battery, ROV, Sub…) providing that the voltage ranges from 24 to 30 Vdc and that the peak accepted current can reach 3A.

User should refer to Hydroptic to get procedure to change settings if voltage under 26 Vdc is provided to UVP5.

The UVP5 Deck Unit provides about 30 Vdc to power the UVP5 when downloading data and allows light tests on deck.

8.2 Power management

The voltage of the internal battery can by checked by :

1) Reading the voltage on the Deck Unit (see below).
2) Reading the voltage via the status messages (V092 messages, see below) sent by the UVP5 via the RS232 link (using a Terminal or the Zooprocess application).
   The Sea Cable ONLY will be connected from the UVP5 to the Deck Unit.

8.2.1 Battery control via the Deck Unit (recent version)

The Power cable must be connected to the UVP5 and the Deck Unit.

The “External Power” switch can be ON or OFF during the charge of the battery.

This version of the Deck Unit permits to visualize the voltage ❷ of the battery (when the power cable is connected and the rear switch of the D.U. is ON ❶) without discharging the battery.

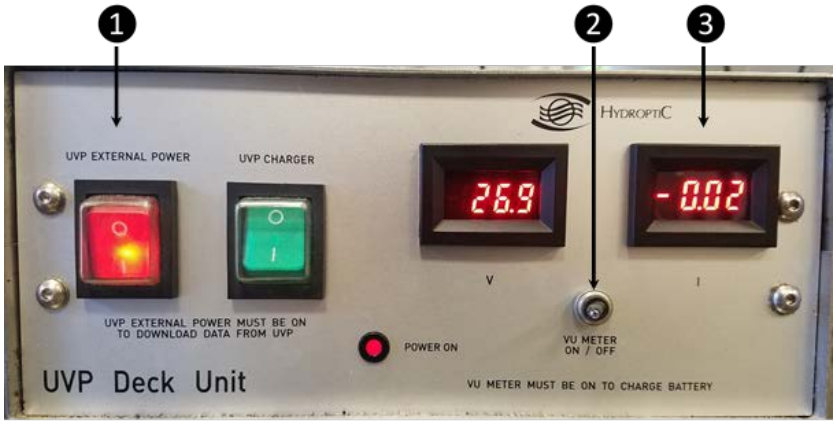


New version deck unit

8.2.2 Battery control via the Deck Unit (old version)

The Power cable must be connected to the UVP5 and the Deck Unit.

The "External Power" switch can be ON or OFF during the charge of the battery.



Old version deck unit

The Battery voltage can be read on (V) meter only if:

- The Vu-meter ❷ SWITCH is ON
- The Amp meter (I) ❸ indicates negative value (battery powering the Volt Meter)

The "External Power" switch can be ON or OFF during the charge of the battery.

Because the control of the battery voltage requires to switch the Vu-meter ON (and power it from the battery), it drains some power from it. It should thus NOT be kept ON **except during the charge of the battery when it is mandatory**.

8.2.3 Voltage control via a terminal (V092 messages)

We use **Motocross.exe** as a terminal but any other terminal application like Teraterm will do it.

The application is located in the USB key under

\UVP5_SNXXX\UVP5_SN_Laptop software\

IMPORTANT NOTE: Battery voltage can be read via the RS232 serial link ONLY when the UVP5 is powered on its internal battery via the POWER SHUNT. In any other case, you will read the Deck Unit power value instead.

The VOLTAGE can be checked via the TERMINAL application or via the ZOOPROCESS checking tool.

1) Serial cable must be plugged into laptop USB and Deck unit ④ (image 8)
2) Start the terminal application. The com port number depends of the laptop used and the USB-RS232 plug.
3) Power UVP5 using the POWER SHUNT
4) The RS232 messages to be received are the following:

Running UVP5 Pilot Application.
You have 5 s to press 'q' if you do not want to launch the program 5 4 3 2 1
Testing DAC WAIT !
V092DATA_-
0002*00179*00179*00022*02734*02277*02058*02070*01732*01861*01406*01838*070
7122726!
V092DATA_-
0002*00178*00178*00023*02719*02572*01975*01683*01879*01630*01776*01889*070
7122726!
V092DATA_-
0002*00178*00178*00023*02726*02635*02154*02209*01823*01907*01408*01880*070
7122730!

The **« V092…! »** lines are generated by the UVP5 controller (mother board), while in supervision mode and enable the system to be monitored continuously. The command line structure is the following:

<V092DATA***ppppp***iiiii*iiiii*ttttt*aaaaa*aaaaa*aaaaa*aaaaa*aaaaa*aaaaa*aaaaa*
aaaaa***mmddhhmmss**!>

Where:

ppppp: Pressure in centi-bars (the value is then divided by ten in order to reach a metric value).

iiiiii: Angle of the system in degrees (the system is by default stabilized in vertical position at 180°).

ttttt: Inside **temperature** in Celsius

aaaaa: **The first aaaaa variable displays the voltage (in mV) of the Lithium-Ion battery** or of the **Deck Unit** depending on whether the power shunt or power cable are connected.

aaaaa : the temperature of the hot side of the cooling device of the camera (HD models)

aaaaa : the temperature of the camera (HD models)

mmddhhmmss: UVP5 internal time and date (does not display year)

8.2.4 Voltage control via Zooprocess

Refer to Zooprocess piloting tools and utilize the “Monitor com port” menu. The voltage is displayed in ImageJ status bar.

8.2.5 Battery STATUS and AUTONOMY

The internal cells enable a variable duration of operation for the UVP5 that may vary depending on the status of the system.

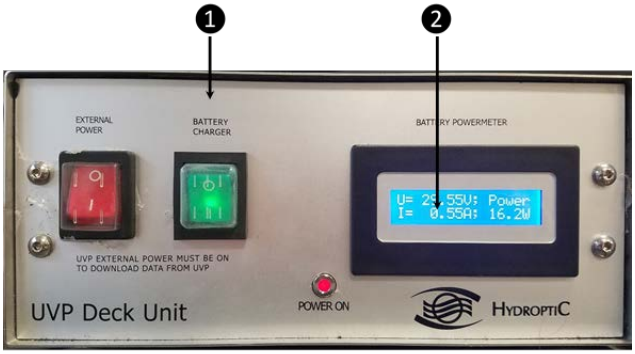
The camera cooling system of UVP5hd models is turned ON with the UVP and may drain power if the outside temperature of the UVP is high.

There are 4 statuses with their own consumption as shown in the table hereunder:

Mode		State	Battery life
Transport	OFF	Off	full
Suspend	Sleep	On	7 Day
Supervision	STBY	On	5 Day
Action	Sequence	On	4 Hours

1) OFF: store UVP5 at voltage ranging between 27 and 28 V (preferable).
2) SUSPEND or SLEEP mode is automatically commuted by UVP5 when
    a. Battery voltage drops below the limit set (23.3V by default), internal mother board temperature limit overpassed (50°C by default) or camera temperature overpassed (25°C by default).
    b. UVP5 is manually turned to sleep mode
    c. Between TIME programmed sequences of image acquisition and process
3) SUPERVISION mode is started when the UVP5 is powered and properly configured to run acquisition sequences in other modes than TIME.
4) ACTION mode imply that UVP5 camera is powered to:
    a. Acquire and process images (LIGHTS are ON too)
    b. Download data
    c. Change settings

8.2.6 Battery CHARGE



New version deck unit



Old version deck unit

The UVP5 battery must be charged when battery drops below 27.5Vdc and you will have performed a profile at sea.

The Battery is fully charged at 29.2 - 29.4Vdc.

The UVP5 will switch to SLEEP mode (not responding to commands, not sending any RS232message) if battery below the limit set (23.3Vdc). A series of STANDBY messages will have to be sent to wake up the system.

It requires 4 hours to charge a fully discharged battery. The charger will automatically stop charging after 4 hours.

PROCEDURE:

1) CONNECT POWER CABLE to UVP5 and Deck UNIT.
2) TURN ON UVP charger green switch ❶ to charge the battery.

The battery is being charged only if Amp meter (I) indicates POSITIVE value ❷ => charge to the battery

8.2.7 Battery monitoring

It is very usefull to monitor the voltage of the battery at the end of each deployment. It will allow to immediately evaluate:

- If the battery needs to be charged
- If the instrument worked (the voltage dropped since previous recording of the voltage)

8.2.8 Battery CHARGE in COLD conditions

In order to get a longer battery life, battery should not be charged when below 10°C.

The internal UVP5 temperature can be monitored in Terminal mode (see V092 messages) or using the Zooprocess pilot tools to monitor COM port. Note that temperature

measurement is accurate only at the moment UVP5 is powered ON. The board temperature increases rapidly biasing the measurement.

If you can store the UVP5 in a warm place and you have lot of time, just wait till the temperature raises 10°C.

If you are in a rush or the UVP5 stays outside in the cold, you can try to warm up the internal temperature turning ON the camera (as for downloading data) before and during charging. Apply all necessary care for that operation.

9 Operating the UVP5

9.1 General information

9.1.1 UVP5 Four main status

The UVP5 has four states of energy consumption which are described below.

9.1.1.1 Power Off for transport or storage

The Power Dummy is connected or the Deck Unit Vu-meter is OFF when connected via the Power cable: no power consumption.

9.1.1.2 Suspend or Sleep Mode

This mode is activated in several cases listed hereunder:

- RS232 command line sent by the user
- Low input voltage detected by the UVP5 at 30 second intervals
- High temperature detected by the UVP5 at 30 second intervals
- High camera temperature detected by the UVP5 at 30 second intervals
- No sequences or time only related sequences are programmed in the system

The power consumption is limited to a few tens of mA when the system is in these modes.

9.1.1.3 Supervision or Standby Mode

This intermediate Mode allows the monitoring of the pressure sensor to be performed by the system since the driver board is supplied and operational. This is the default Mode for the UVP5 when an I/O, Pressure or Manual sequence is programmed and UVP5 turned ON (via Power shunt or Deck Unit). The V092 are continuously displayed when in this mode.

The Standby mode drains the same energy. It permits to change UVP5 settings.

9.1.1.4 Action Mode

The camera, is powered for a programmed sequence or a data download RS232 command line. Light are ON if in recording mode.

The recording autonomy is approximately 4.5 hours depending on temperature and battery charge.

9.1.1.5 Thermal Issues Warning

The UVP5 needs to be immersed or cooled down in the event of the outside temperature being above 25°C and sunlight warming the UVP5 pressure case.

9.1.2 Computer tools

ALL applications are already installed on your laptop by Hydroptic. You may anyhow have to update them to get benefit of the latest improvements.

The piloting software and drivers are located in the USB key.

Updates can be found on the www.hydroptic.com and PIQv websites.

9.1.2.1 USB-RS232 and Com Port

Laptops do not have serial port anymore (except industrial ones). The new Deck Unit includes a serial-USB converter while we provide a RS232-USB cable adaptor with the serial cable for older versions of the deck unit.

Both adaptors come with a driver. Install the driver first.
Location: \Serial port to USB adaptor driver

Note: the USB cable or the USB-RS232 adapter must always be connected to the same USB port of your computer to avoid changing the COM port number in the terminal application or in Zooprocess.

Once the driver is installed and the adaptor connected to the laptop, go under your laptop /hardware device manager/port com and write down the port number. You will use it when configuring motocross and Zooprocess.

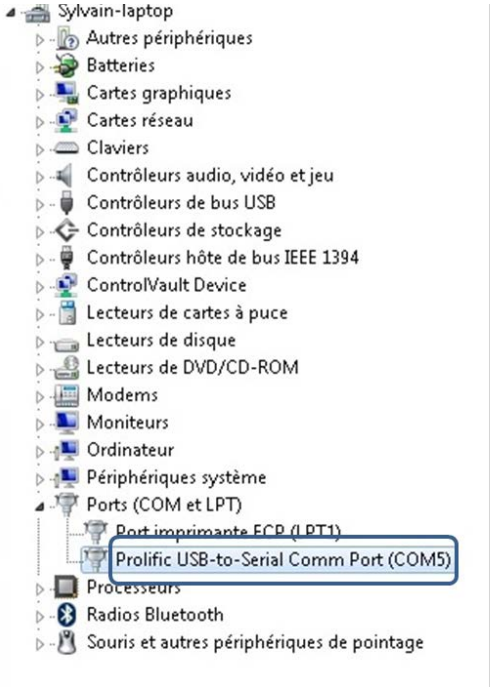


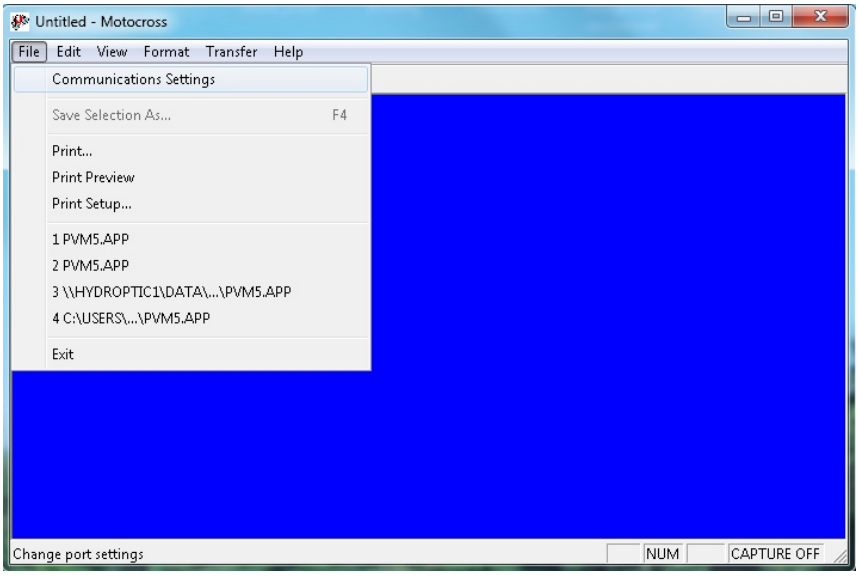
Image 9B

9.1.2.2 Terminal piloting

Motocross is a terminal application provided with the UVP but any other terminal will work. Teraterm.exe works well on W7 and W10 64bit operating system and can be a useful solution when the serial com number is higher than 9.

- In case you have to re-install Motocross, just copy the application from the USB key onto the desktop of your PC.

- Enter the correct port under the *communication settings* tab as well as the baud rate which is 9600. Keep all other as default. NOTE that MOTOCROSS accepts COM port numbers below 10 only. Com port number can be changed in the device manager tools.



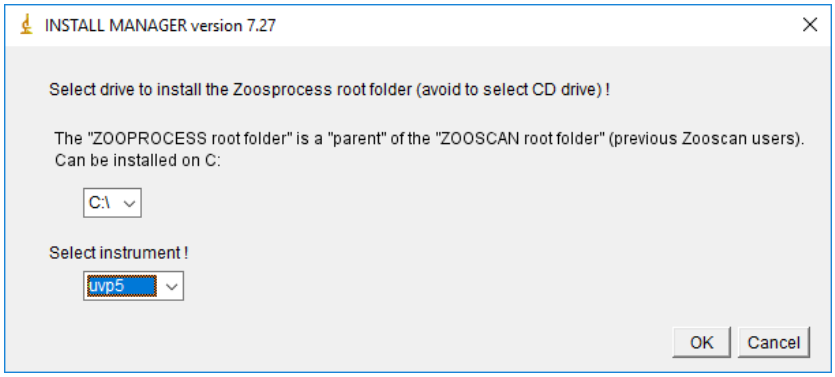
9.1.2.3 ImageJ and Zooprocess

ImageJ is a java-based image processing program and Zooprocess a suite of routines in ImageJ macro language. Both are already installed on the UVP5 computer.

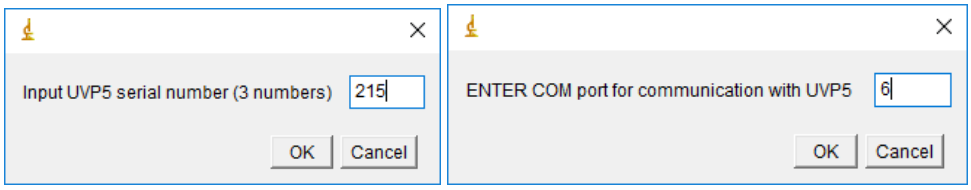
In case you have to re-install these applications, please follow carefully the steps explained in the *ZooProcess_Setup_Manual_vxxx* located on the memory stick and preferably updated from the PIQv website.

9.1.2.4 Default settings of Zooprocess for UVP5

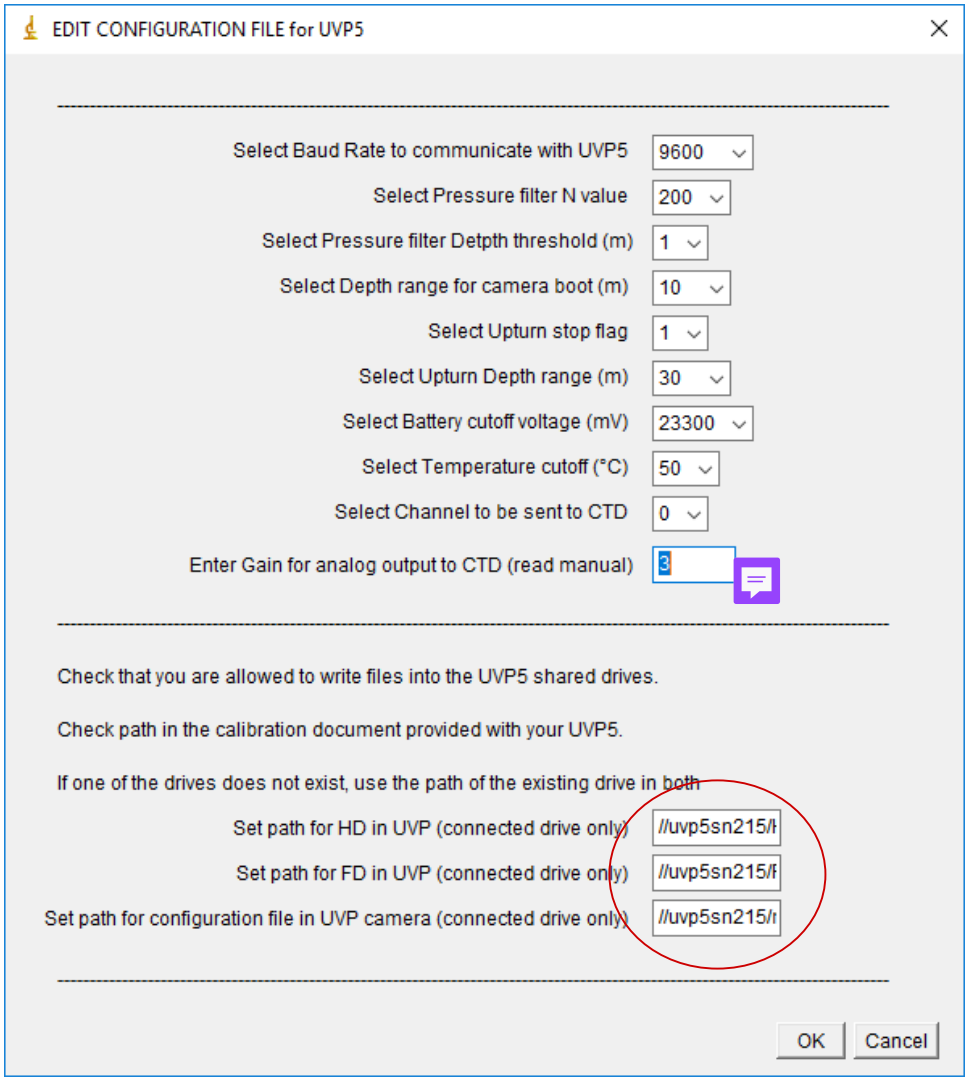
The very first time you will start Zooprocess, you will have to set the UVP5 as your working instrument.



You will be asked to precise the SERIAL NUMBER of your instrument. Please carefully document this field. You will also be asked to choose the serial port number if you have to communicate with the instrument to pilot it. Check this port as detailed above.



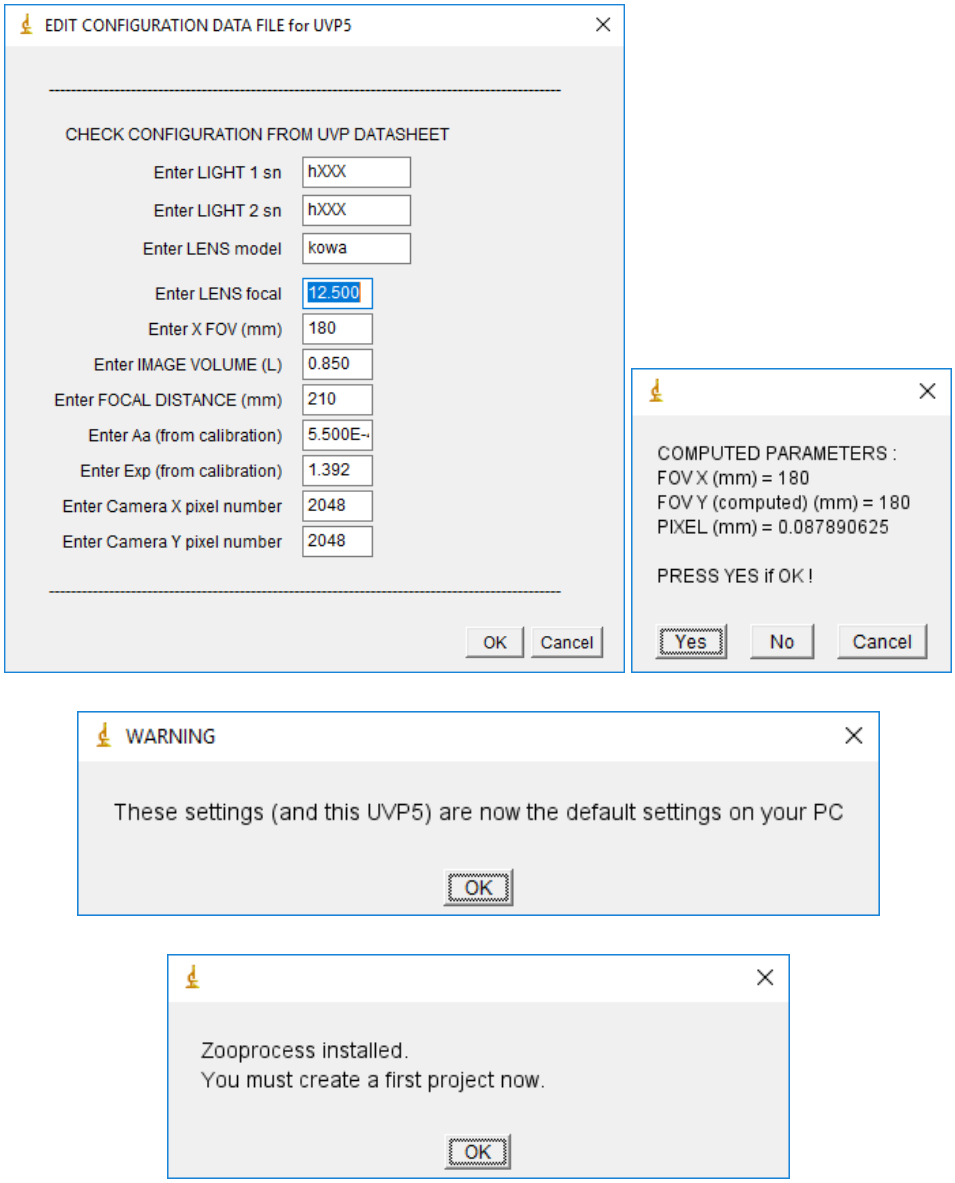
You will also be asked to check some parameters in some menus. These parameters are now downloaded from the instrument where the correct parameters have been documented after its calibration. **You thus only have to care about the serial number of your instrument in the configuration paths at the bottom of this screen.**



9.1.2.5 UVP5 hardware settings

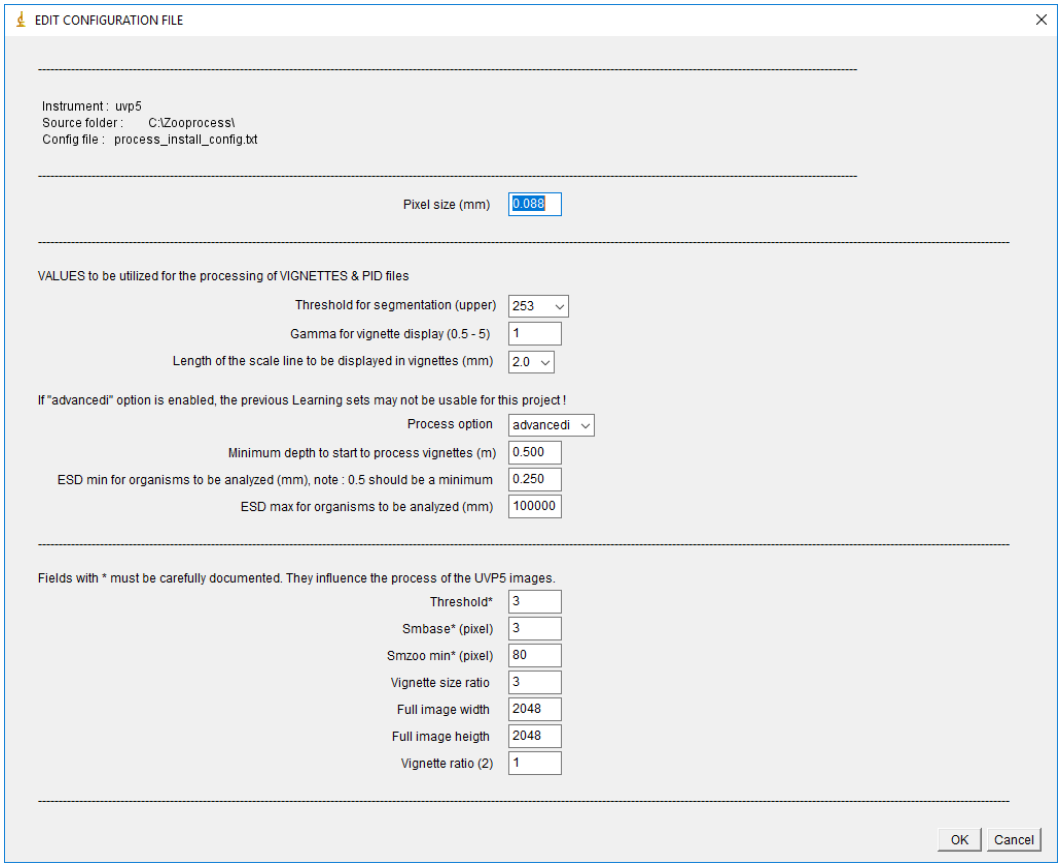
The next step is to document the configuration DATA for your instrument.

These settings will be downloaded and UPDATED from the UVP5 everytime you will create a new project. You thus do not have to worry about these parameters at that stage.

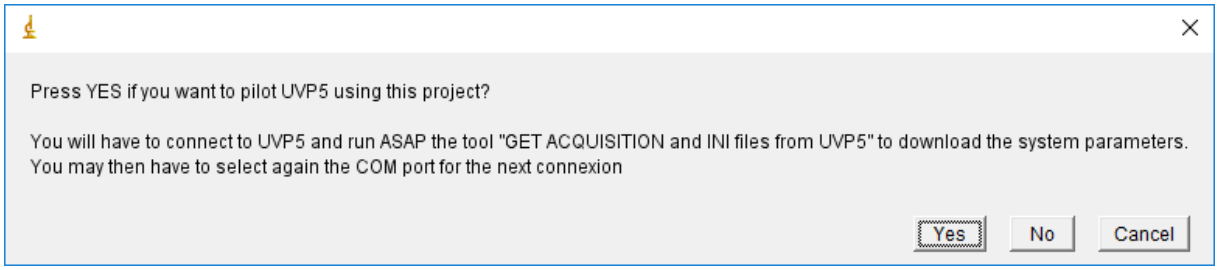


You will then be asked to create a first project, select again the serial number of your instrument and check image process parameters. **These parameters will also be downloaded from the instrument the first time you will enter the PILOT tools. You thus do not have to care about them except if you do not want to PILOT the UVP5 from this project.**

Please carefully fill in the cruise information when asked. This information will follow the data during all their life in Ecotaxa.



The last screen indicates that you will have to connect to the UVP5 to get the files when you will run the PILOT tools for the first time in this project.



9.1.2.6 Project arrangement

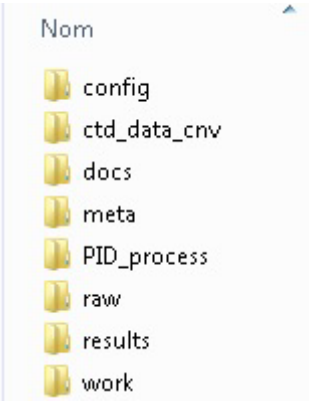
Your UVP5 has been qualified and calibrated prior delivery. The results of these operations are provided with your system. Calibration and configuration files for your UVP5 are located in the folder **Calibration_report_UVP5_snXXX_YYYYMMDD.pdf.**

The proper settings have been set **in your instrument** and will be downloaded every time you will create a new folder and run the proper PILOT tools.

The Zooprocess manual describes well the folder arrangement of the UVP5 projects. This manual emphasis on the tools related to piloting of the UVP5.

The project folder is composed of 8 sub-folders as shown hereunder and summarized in the next paragraphs.

Each new created project under Zooprocess will always be organized with the same tree structure. We recommend creating a new project for **each cruise** you will perform. Run Zooprocess for that purpose!



9.1.2.6.1 Config folders

The first folder named **config** folder contains 4 files plus the **UVP5_settings** subfolder utilized by Zooprocess to pilot the UVP5.

Apart from the "profiler.ini file, the three other files are requested only for the Zooprocess processing of the profiles.

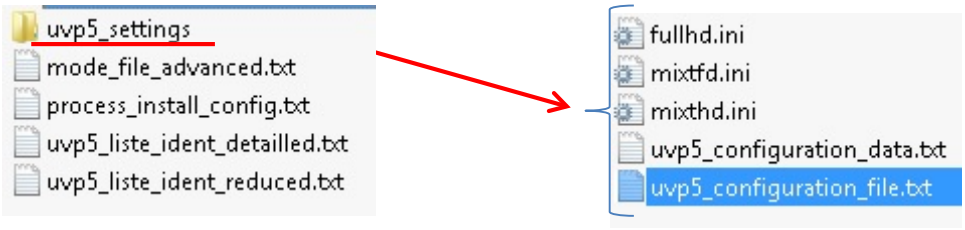
- The **UVP5_configuration_data.txt** contains originals settings values necessary to post-process images with Zooprocess.



- The **mode_file_xxxxx.txt** set the Zooprocess running mode to USER or ADVANCED.
- The **cruise_info.txt** file which is filled when you create a project. This information is essential as they will follow the data in Ecotaxa.

9.1.2.6.2 UVP5 settings subfolder

The **config** folder contains also the **UVP5_settings** folder which has at least 5 files as shown hereafter. These files are already set in the UVP5's camera memory. **These files are downloaded from the instrument at the first start of the PILOT tools in a project.** They are to be kept on your computer for information and processing purposes.



UVP5 camera files (see below):

- **Fullhd.ini** is chosen for images to be processed and save. Images will be saved onto the HD disk.
- **Mixtfd.ini** is chosen for images to be saved and processed. Images will be saved on the FD disk.
- **Mixthd.ini**, same as Mixtfd.ini but images are saved on the HD disk.

GENERAL configuration file (specific to your S/N):

- **UVP5_configuration_data.txt** includes parameters for the field of view as well as the lens settings, the image volume and size plus the calibration parameters.
- **UVP5_configuration_file.txt** contains the port com and baud rate that should match the computer com port in order to communicate with the UVP controller by the mean either of a terminal (Motocross) or Zooprocess. (ImageJ). Other parameters are included (image 10). For instance: Battery voltage and temperature threshold safety features.

```
PilotBR= 9600
comport= 3
databits= 8
stopbits= 1
parity= NO
Pressure_Filter_Nb= 200
Presure_Filter_Thres= 1
Pressure_Pre_Pos= 10
Pmin= 0
DeltaP_Upturn_Flag= 30
Upturn_Flag= 0
Battery_Threshold= 25200
Temperature_Threshold= 50
CNA_Ind= 0
CNA_s= 20
CNA_o= 0
hd_path= //uvp5sn006/HD
fd_path= //uvp5sn006/FD
modfile_path= //uvp5sn006/FD
```

The configuration file can be edited with a text editor or with zooprocess. (See above)

9.1.2.6.3 Raw folder

The raw folder contains the profiles folder downloaded from UVP5 after completion of profiles (see chapter 7.2.5).

Each HDRYYYYMMDDHHMMSS sub folder contains at least

a) *.HDR file : copy of the image acquisition file utilized by the camera (mixtfd, mixthd, fulhd)
b) *.DAT : N°, date, depth, sensor readout, gross image data for each of the image
c) *.BRU : individual measurements for each of the detected object
d) *.TSv files : synthetic files of the data for some size classes of particles
e) Image files : their number and format depends on the settings.

9.1.2.6.4 Work folder

This folder will contain the folders of the processed samples/profiles/images.

Each of them will contain different files according to the selected process and instrument.

9.1.2.6.5 Meta folder

Contains a unique file documenting the metadata of all samples/profiles will be created and completed there. This file must not be deleted and user must be careful in attempting to manually modify it. It is preferable to use the EDIT & MODIFY METADATA tool for that purpose!

9.1.2.6.6 CTD DATA CNV folder

Place to store the CTD data files.

9.1.2.6.7 Results folder

This folder will be automatically completed by the processed files and graphs.

9.1.2.6.8 Pilot check folder

The logs of the PILOT tools will all be saved there allowing to later diagnose possible troubles of the instrument.

9.2 Field operation

9.2.1 Introduction

The UVP5 will in most cases be attached in or out the CTD frame. The UVP operation will thus be performed in relation with the CTD operators, the ship watchman and the winch man. It is very important that the UVP operator communicate with these people prior to UVP deployments to explain them the way the UVP is programmed and will work.

The UVP5 can start and stop acquisition in four modes:

1) Pressure : uses the embedded pressure sensor and a specific logic
2) I/O : uses a switch (I/O shunt) to start and stop sequences
3) Manual : RS232 commands are sent by user (or piloting device)

4) Time : UVP5 is started and stopped at user pre-set time

Any type of UVP5 acquisition can be manually stopped by RS232 command. It must not be stopped by removing power (as for any desktop computer!).

The programming of the UVP5 permits to define the acquisition mode and the starting/ending protocol.

9.2.2 Acquisition modes

UVP5 is provided with three defaults acquisition modes: mixtfd, mixthd and fullhd. These modes define how the instrument will acquire and process the images when acquisition is started.

All other image acquisition and processing parameters are the same and coherent with those described in the provided calibration report and project. **Any change may result in a need for additional calibration procedure**.

UVP5 permits four image acquisition and process options (even if only three available as standard) :

1) Save only: records images only, no process in-situ. The Zooprocess post processing of the images is mandatory.
2) Process only: the images are processed but there is no recording of any image, no zooplankton identification possible. Highest acquisition frequency.
3) Process and save (fullhd.ini): process and save all images on HD. It requires lot of space and drops down acquisition rate to 3 Hz on UVP5std models.
4) Save and process (mixtfd.ini & mixthd.ini): process all images and save only images or ROIs of objects to be later identified (above a size limit). Works on FD or HD. The fast and most efficient mode (saves disk space).

The UVP5std is equipped with two memory drives:

5) FD: flash drive containing also the OS of the UVP5 camera. It allows the fastest acquisition rates on UVP5std buts its size (13 Gb) may limit the number of casts in case you save many images.
6) HD: USB2 drive which can be internal or external (option). Higher capacity but lower speed.

The UVP5hd is equipped with one 100Go SSD drive. The data can be stored either on the FD or the HD folders without any difference in performance.

We recommend using the mixtfd.ini configuration which provides the highest acquisition rates and keep only (default) ROIs of objects to be later classified.

The fullhd mode provides the possibility to later re-process images both for particles and Zooplankton with different settings (threshold, size limit). Useful if you suspect the water to be too turbid for the default settings provided with the calibration.

9.2.3 Starting/Ending Protocols

9.2.3.1 Pressure Protocol

This mode is utilized in most cases when UVP5 is attached with a CTD for vertical profiles.

The user must carefully respect the indicated protocol using this **very powerful mode of starting sequences**. This mode saves:

1) Battery
2) Memory
3) Download time

This mode is utilized for SINGLE profile unless user accepts to redo the starting procedure between profiles (no need to have UVP5 back on deck between casts).

Although the measures are in deci-bars, the depth via the metrical unit will be used in this manual.

User must follow the sketch below which corresponds to the default recommended parameters:

- START criteria : "**D**" for descending
- START depth : 15 meters
- Prepositionning dP : 8 meters
- STOP criteria : "**D**" for descending
- STOP depth: 6000 meters (thus not activated !)
- UPTURN flag : activated to automatically stop UVP acquisition
- UPTURN dP : 30 meters (more than the CTD rinsing depth)

The above fields will be documented either in TERMINAL mode (see RS232 commands below) or using Zooprocess "READ / PROGRAM sequences" menu which will assist you.

This sequence takes advantage of the CTD "rosette" 20 meters soaking prior to diving the system into the water so as to automatically start-up the image acquisition. The operations can be performed without the presence of the UVP operator if the CTD team is well aware of the sequence of operation and the recommendations below. The CTD operator will only have to disconnect the Power cable and place the Power Shunt prior to deployment and back when on deck (if flashes are OFF).

The start-up will take place at 22 meters under the surface when in Standby for soaking. UVP5 camera will enter pre-positioning at 7 meters (ie dP = 8 meters, if descending speed is sufficient) when descending to 22 meters.

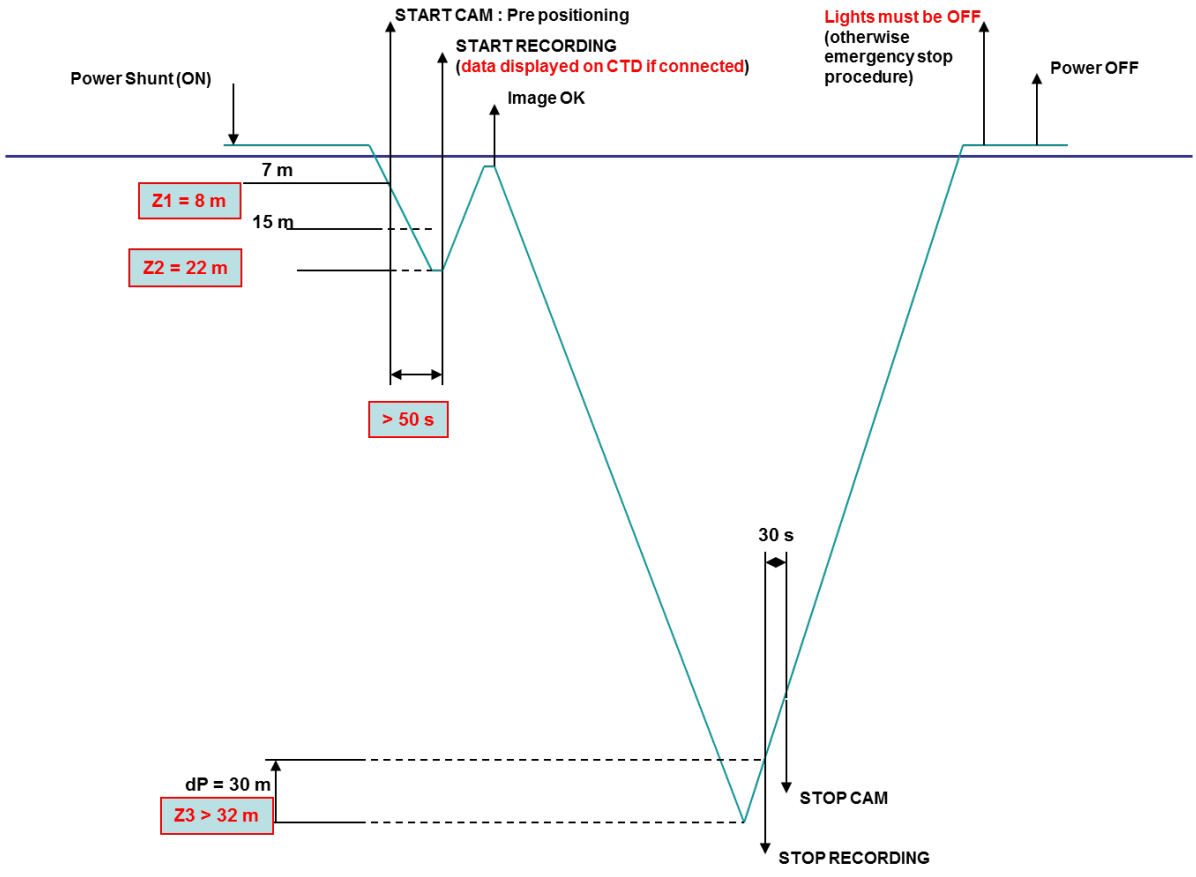
- If the UVP5 analog output is connected to the CTD, the 22 meters standby can end when CTD operator detects signal on the UVP5 analog input (Voltage).
- If the UVP5 is not connected to any CTD, you must wait more than 80 seconds after it passes 7 meters to allow rising up the UVP5 under the surface and start the profile.

The end value of 6000m is never reached since the interrupt is triggered by the return to surface flag whatever the depth once higher than 35 meters.

The descent speed needs to be at least of 0.3 m/s when reaching the depth of 10 meters.

Recommendations:

a) If the UVP does not start (not data output on CTD screen after UVP5 80 seconds boot delay) at 22 meters, rise up UVP below the surface and redo procedure descending faster if possible.
b) If the UVP red flashes not visible (not applicable in daylight) from the surface before the start of the profile (Image OK step), redo the procedure.
c) If a profile needs to be aborted for any reason (CTD failure), descent the system below 35 meters in order to take benefit of the automatic UPTURN stop.
d) If UVP5 still flashing when back on deck, connect DATA cable and stop UVP5 by RS232 command line or by Zooprocess. **Do not remove Power shunt till you properly stop acquisition and wait more than 30 seconds afterward.**



The Power Shunt can be connected on the Power Extend in advance (knowing that the UVPhd models will drain some power to coll its camera).

In case the UVP fails to start using this rinsing procedure and the initial descent speed cannot be increased, an alternative solution which requires the presence of the UVP operator is to keep the above settings of the UVP but manually start acquisition on deck (Power Shunt in place and Sea Cable connected) just before deployment. The UVP5 will start record images on deck or during deployment and normally ends when rising up by 30m. This procedure permits to save energy and memory but requires the presence of the operator and the use of the Sea Cable for each deployment.

9.2.3.2 I/O protocol

The acquisition sequence can be manually started using the I/O shunt connector (yellow). In the event of the connector being linked to a driver system (ROV, AUV), the shunt can be replaced by a relay and enable easy sequence starts.

This mode can be selected for classical vertical profiles when it is required to manually start the sequence prior to launching and interrupt it by the same means upon return to surface. **It is very useful for shallow profiles** when 35m depth cannot be reached to allow upturn flag to stop UVP5 on ascent.

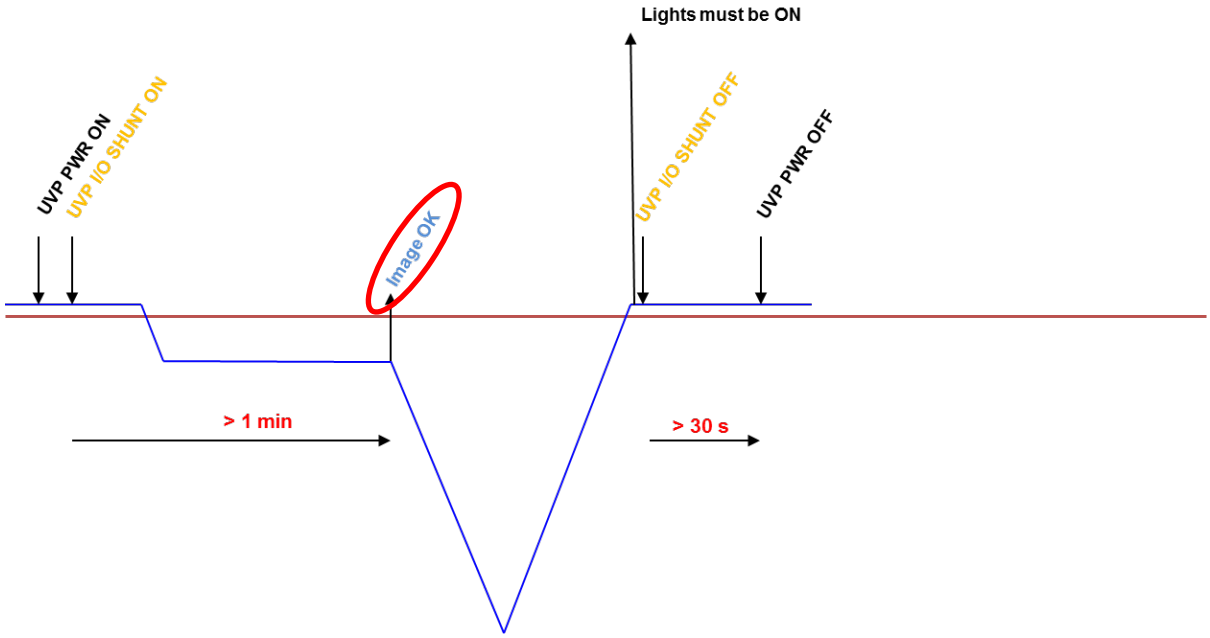
The mode is very easy to manage but nevertheless requires the presence of an operator for each launching.

It allows easy YOYO casts as described in the graph but permits also single casts.

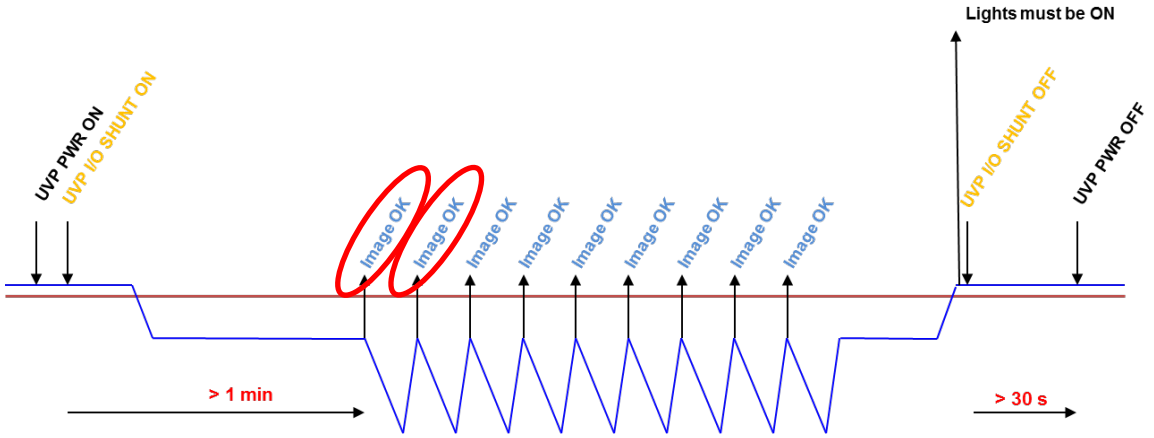
This mode is less efficient than the depth mode as it:

1) drains more power as it records descent AND ascent profiles and also makes images on deck
2) uses more memory for the same reason
3) requires more download time for the same reason

9.2.3.2.1 Example of SINGLE I/O cast



9.2.3.2.2 Example of YOYO I/O casts



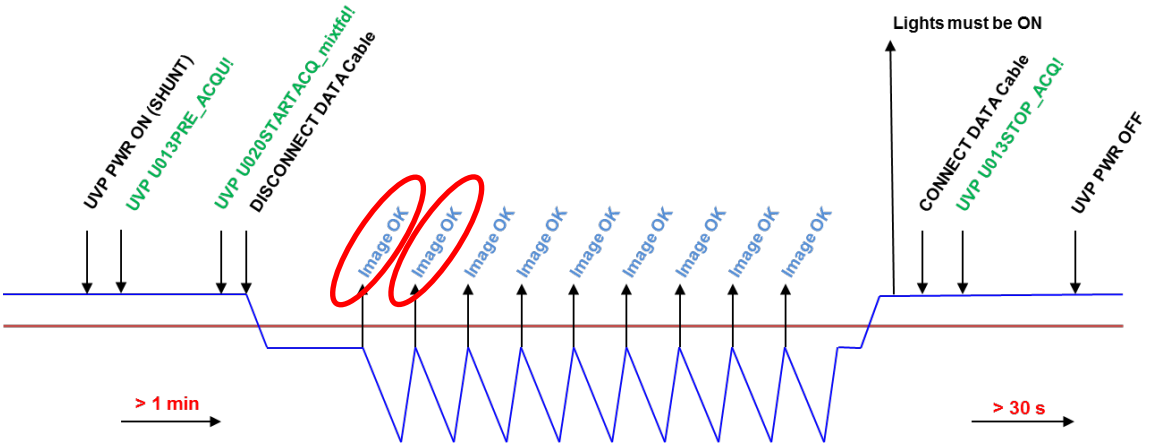
9.2.3.3 RS232 commands protocol

The acquisition sequence can be manually started via the usage of an RS232 command in Terminal or Zooprocess.

This mode requires the action of an operator before and after the launching of the UVP5. It is more demanding than the I/O mode as the Data cable will have to be disconnected before launching the UVP5.

It is anyhow very similar to the I/O mode. It can also be utilized in conjunction with the Depth mode in case of difficulties to start the UVP during the rinsing procedure.

This mode can be run either in a Terminal or in Zooprocess using the dedicated tool.



9.2.3.4 TIME protocol

Starting and ending times are entered in the UVP5. UVP5 goes into SLEEP mode between recording sequences which start and stop at preset times if UVP5 is powered.

9.2.3.5 Emergency stop procedure

In case the UVP is programmed in I/O mode or RS232 modes, it is normal that the UVP flashes (and keep recording) when back on deck. In I/O, you will remove the I/O shunt according to the I/O procedure. In all other modes but Time mode, you will **connect the Sea Cable** and manually stop the acquisition before removing the Power Shunt.

If you use a Terminal application (see below), just copy and paste the command **U013STOP_ACQ!**.

If you run Zooprocess, close ImageJ and start it again. Select the “UVP5 piloting tool”. The application will detect that the UVP is in acquisition and will offer to stop the acquisition.

9.2.3.6 Additional information on Pressure protocol

In Pressure mode, the user defines whether the sequence start-up is performed while in ascending criteria (A) or descending criteria (D) for a given depth.

Two parameters allow the UVP5HD to detect whether it is ascending or descending. This validation allows unnecessary start-ups not to be triggered due to pressure variations caused by the roll of the ship.

The instrument makes an average calculation of the nnnn successive pressure values and compares them to the following nnnn values.

- Should the delta be positive and the absolute value higher than the pre-defined ssss pressure value parameter, the system is considered as is in descending mode

- Should the delta be negative and the absolute value higher than the pre-defined ssss pressure value parameter, the system is considered as is in ascending (back to surface) mode.

The parameters are defined via the <**U024SET_PFILT_nnnn_ssss!**> command line where nnnn represents the number of successive pressure values (default value : 200) and ssss corresponding to the delta(default value : 1) in pressure (measured in dec-bars).

The definition and linked adjustments of the mode currently in place (ascending or descending) depends on the frequency of the pressure measures. The frequency not being clearly defined; adjustments will be made as follows:

- The higher the ssss value the closer the control will be on the system
- The higher the nnnn/ssss ratio, the lower required speed of the system will be.

9.2.3.6.1 Start-up in Descending Mode

Whenever a pressure defined sequence has been validated by the UVP5, the pressure is continuously measured and the A/D (Ascending/Descending) test activated by the system.

The pre-positioning will start automatically when the descending criteria has been validated and the depth is higher than that pre-defined value minus the pre-positioning interval.

At the end or the pre-positioning, the system automatically starts up if the depth is equal or higher to that defined in the sequence or the system will wait for the pre-defined pressure value to be reached.

9.2.3.6.2 Start-up in Ascending Mode:

Whenever a pressure defined sequence has been validated by the UVP5, the pressure is continuously measured and the A/D (Ascending/Descending) test activated by the system.

The pre-positioning will start automatically when the ascending mode has been validated and the depth is lower than that pre-defined value plus the pre-positioning interval.

At the end or the pre-positioning, the system automatically starts up if the depth is equal or lower to that defined in the sequence or the system will wait for the pre-defined pressure value to be reached.

9.2.3.6.3 Descending Mode Interrupt:

Once the depth is higher than the pre-defined value and that the system is in descending mode the sequence will be immediately stopped.

9.2.3.6.4 Ascending Mode Interrupt:

Once the depth is lower than the pre-defined value and that the system is in ascending mode the sequence will be immediately stopped.

9.2.3.6.5 Pressure Value Start-up and Pre-Positioning Value:

In order to prepare the acquisition start-up prior to the defined depth value being reached, a pre-defined distance is integrated via the <**U019SET_PPREP_pppp!**> command line where pppp stands for the interval in metres.

9.2.4 Piloting the UVP5 using ZOOPROCESS

9.2.4.1 Introduction

Zooprocess uses the same set of commands as if they were entered one by one under Motocross terminal. Nevertheless, Zooprocess is user friendly and no mistake can be made thanks to its GUI.

You can access the piloting tools after entering (or creating) your project selecting the "**UVP5 Piloting tool**" menu.

All information is displayed

1) In the LOG window
2) In the ImageJ status bar
3) In specific windows

Be careful when powering the camera. **Use any convenient method to avoid switching UVP5 OFF when camera is ON for any purpose**. Use stickies or any reminder if you have to move in order to avoid someone else making a mistake. Other efficient precaution is to avoid running any other application on the PC than Zooprocess in order to avoid blinding precaution messages.

USER mode provides all necessary tools for daily operation of the UVP5 and prevents making errors using the advanced tools. **It is thus recommended to run the Piloting tools in USER mode only.**

9.2.4.2 Monitoring tool

Zooprocess provides a tool to monitor and display UVP5 messages in real time when connected to a serial link (SBE option on CTD SBE911). This tool does not allow to send any command to UVP5 which will be started using either I/O shunt, Time or Depth method.

9.2.4.3 List of TOOLS available in USER mode

a) TEST & MONITOR COM PORT: displays received messages (V092 type and other) and data into the Status bar of ImageJ.
    a. UVP date and time
    b. UVP voltage
    c. UVP angle
    d. UVP depth
    e. UVP internal temperature
b) Synchronize UVP5 time with Computer time
c) DOWNLOAD DATA from UVP5: starts camera to download data from UVP5
d) DELETE DATA in UVP5: uses the RESET terminal command to delete data after powering the camera. Note that Data can also be manually deleted via the network when camera is started (more reliable when many casts stay in UVP drives and need to be deleted).
e) READ / PROGRAM sequences: Check actual UVP5 programmed sequence and allow changing it to DEPTH, I/O, Manual or Time. This tool is much safer than

the Terminal mode as it will also set the additional parameters to secure the mode.

f) START ACQUISITION: starts acquisition in Manual mode. C110 messages will be read and essential parameters displayed in the ImageJ status bar. A graph of the particle concentration, size and grey level can be displayed using Depth or Time as y axis.
g) GET ACQUISITION and INI files from UVP5: starts UVP5 camera and copy image acquisition files (mixtfd, mixthd, fullhd…) from UVP5 camera to the "settings" folder of the project.
h) EMERGENCY STOP : stop acquisition if UVP5 was not automatically stopped
i) WAKE UP sleeping UVP5: helps to wake up not responding UVP5
j) TURN ON UVP5 light (test / alignment): permits to turn ON lights and make them flashing during 30s for check or alignment (not available on all instruments).
k) SWITCH mode (advanced/user): allow to access all menus

9.2.4.4 Additional TOOLS available in ADVANCED mode only

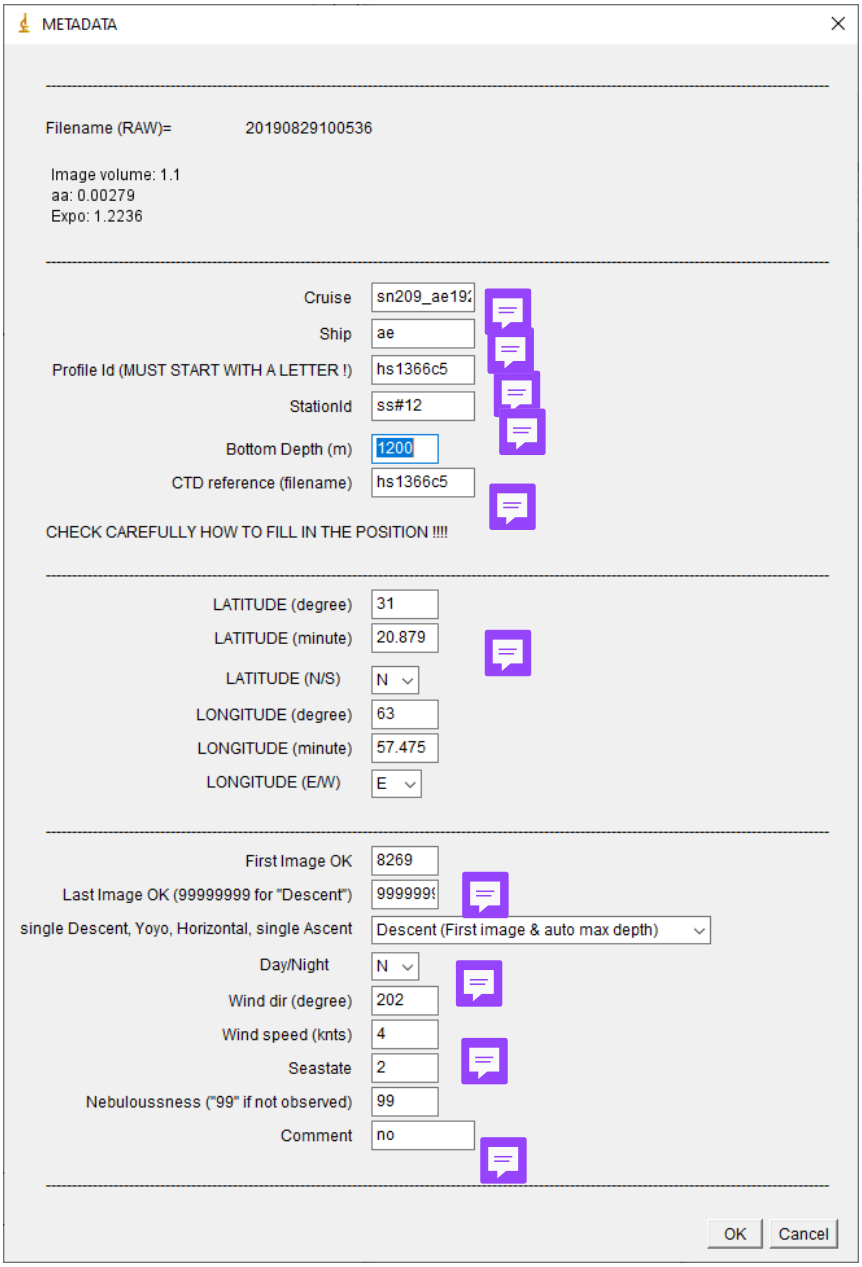
We consider that these tools permit to change many settings and that some modifications may result in UVP5 running problems or need for additional calibration. Great care should be applied and Hydroptic asked for advises if any doubt.

a) EDIT ACQUISITION files: allows to modify those files and to create others.
b) EDIT ProfileurVideo ini file: allows editing the UVP5 internal configuration file (Profiler.ini) which has been previously downloaded to your project.
c) SEND ACQUISITION and INI files to UVP5: the files you created/modified will be sent to UVP5. This action is mandatory if you want to program sequences using these files. Note that the profiler.ini file will be sent but UVP5 will not take the new parameters (to be solved in the next).
d) TURN ON UVP5 camera: allows to manually power ON the camera to manually download or erase data via the network.
e) TURN OFF UVP5 camera: can be helpful.
f) EDIT Pilot board Configuration file: permits to change UVP5 mother board settings such as Voltage and Temperature limits.
g) SEND configuration to pilot board: permits to send those modified parameters to UVP5 mother board
h) Set GAIN for ANALOG output: permits to adjust the analog output of the UVP5.

9.2.5 Basic processing after downloading data from UVP5

9.2.5.1 Fill in metadata

The metadata must be filled first in order to allow processing the downloaded data.



Cruise :	should not be changed, issued from the project name !
Profile Id :	unique name associated with the profile, can include station name. There can be many Profile Id performed during a unique station. This name CANNOT be later changed.
Station Id :	Name of the station where many profiles can be performed
CTD reference name :	can help associating CTD data to UVP5 data
Latitude and Longitude :	The latest version of Zooprocess permits to fill in the lat/lon information on an easy way.
First Image OK :	to be read in the DAT file of the profile (HDRYYYYMMDDHHMMSS_XXX.dat). This number will be the start of the descent (see UVP5 pilot tool manual).
Last Image :	to be modified if the I/O mode has been selected or if multiple profiles have been performed as a yoyo during a unique acquisition sequence. Kept set to 99999 if not to be set. Par défaut

c'est 9.9999E10 qui ne passe. Dire qu'il faut se référer au manuek Ecopart pour modif en 999999.

Descent/Yoyo/Ho : to be set to Yoyo if a single sequence contains multiple profiles. CANNOT be later modified. In that case, all profile metadata of a unique sequence must be filled in the same metadata filling session. Descent permits Zooprocess to automatically detect deepest image of the profile. Other permits all options using unique set of First and Last images for a profile.

Image volume, aa, Exp : parameters provided in the documentation of the instrument and filled when you created the project. They should not be modified (available only in editing mode)

The "FIRST image OK" and optional "LAST image OK" will be read from the DAT files. Zooprocess try to set automatically the First Image OK and displays a graph indicating the detected values. The detected value is pre-set in the metadata fill-in window. User can also check in DAT files if the values are not well selected.



9.2.5.2 Process DAT BRU PID & VIGNETTES

This tool will format the BRU and DAT files for easier use in EcoPart.

It will also process the PID files for computer assisted sorting of Zooplankton and the TSV tables for Ecotaxa.

Note that if you only want to work on particulate matter (LPM), you do not need to process PID and Vignettes immediately as it is a time demanding process.

The DAT and BRU process provides also some plots of the data issued from the real time embedded image analysis. These plots are edited for the 0-200m depth range and for the full profile depth range. They allow to QC the instrument functioning and the data quality.

These plots can also be created directly from the main Zooprocess menu allowing to set detailed parameters (depth, smoothing….).

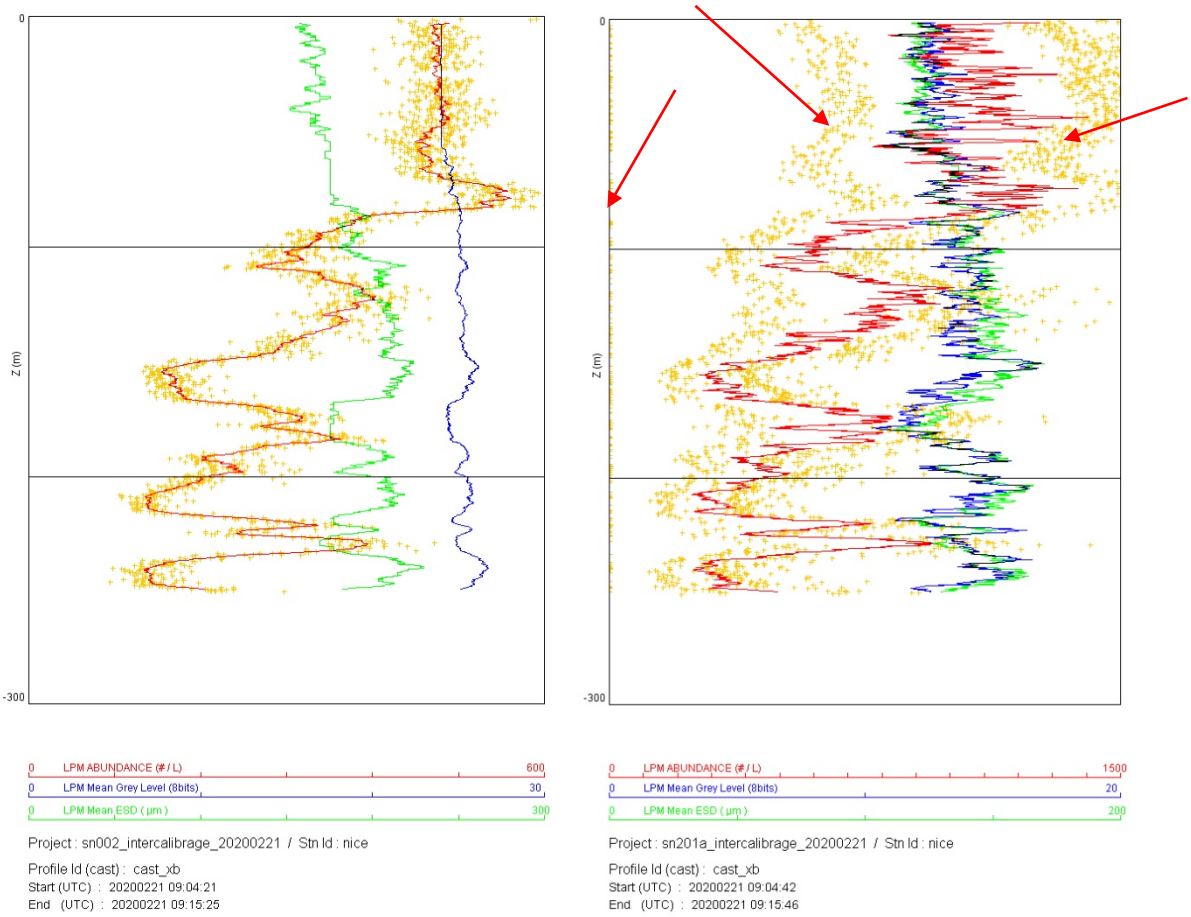
9.2.5.3 Control of the data quality using the plots.

Each of plots created by the DAT and BRU process tool or by the specific menu are useful to check the data quality.

The abundances, gery level and particle sizes are by default averaged on 20 images in order to avoid noisy profiles.

9.2.5.3.1 Plot of LPM grey level and mean size

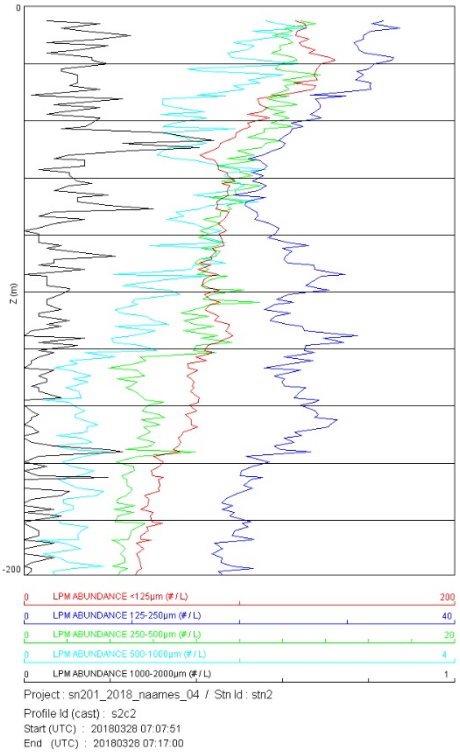
This first plot permits to visualize the most important data from a profile. The RAW abundances (per image) are also displayed in ORANGE. It permits to be sure that both LIGHTS are working well.



The left graph shows that the mean abundance of particle (RED) is centered on the raw image abundances indicating a good functioning of both lights.

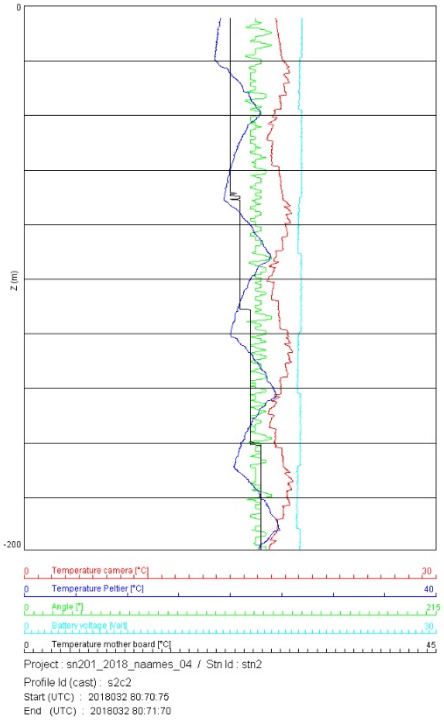
The right graph shows many orange crosses on the left Y axes indicating that both lights were sometimes not functioning. The fact that the red mean profile is not centered in the cloud of orange crosses indicates that one of the two light quite often did not work. The data points on the right of the mean profile are the only correct data (the two lights are flashing).

9.2.5.3.2 Plot of LPM detailed size classes (UVP5hd only)



It is very normal to observe a difference in the shape of the profiles of the different size classes.

9.2.5.3.3 Functioning checks



This graph shows the cooling of the UVP5hd camera during the profile (red) and the temperature of the hot side of the cooling device. The graph also displays the mother board temperature, the UVP battery voltage and the vertical angle (oscillations) of the instrument during the profile.

There is no measurement of the camera temperature and no Peltier on UVP5std.

9.2.5.4 DAT files description

The DAT file contains metadata associated with images acquired and processed by the UVP in situ or post processed by Zooprocess.

This DAT file is later simplified and concatenated in a unique file with no header line and containing the same fields. This final "cleaned" file is saved both into the profile folder (work subfolder) and in the Results folder. It is later utilized in EcoPart to process the per class abundances of particles and plankton.

Example of the first two lines: different fields are indicated using different colors corresponding to the different 4 sections of the line

- index;image;sensor data;nb blobs P-G;mean area P-G;mean grey P-G;nb blobs G;mean grey G;

- 1;20140625160226_000;00010*00205*00205*00030*02815*02181*00759*00865*00490*00694*00774*00796!; 2051;2;22;3;43;

Detailed description:

- **Index: 1** => the image number, starting at 1 in the RAW dat files.
- **Image: 20140625160226_000** => corresponding image acquisition time (YYYYMMDDHHMMSS_ms).
- **sensor data: 00010*00205*00205*00030*02815*02181*00759*00865*00490*00694*00774*00796!** => UVP sensor data

*ppppp*iiiii*iiiii*ttttt*aaaaa*aaaaa*aaaaa*aaaaa*aaaaa*aaaaa*aaaaa*aaaaa

  - **ppppp**: Pressure in centi-bars (the value is then divided by ten in order to reach a metric value).
  - **iiiiii**: Angle of the system in degrees (the system is by default stabilized in vertical position at 180°).
  - **ttttt**: Inside mother board **temperature** in Celsius degree
  - **aaaaa**: **The first aaaaa variable displays the voltage (in mV) of the Lithium-Ion battery** or of the **Deck Unit** depending on whether the power shunt or power cable are connected.
  - **aaaaa** : the temperature of the hot side of the cooling device of the camera (HD models)
  - **aaaaa** : the temperature of the camera (HD models)
- **nb blobs P-G;mean area P-G;mean grey P-G;nb blobs G;mean grey G: 2051;2;22;3;43** => summary of measurements recorded for that image in the BRU file. Here we have:
  - **nb blobs P-G: 2051** objects measuring more than SMbase and less than SMzoo (see modfile related chapter)
  - **mean area P-G: mean area P-G: 2:** average area in pixels of these "small" objects

- **mean grey P-G: 22:** mean grey level (8 bits scale) of these objects
- **nb blobs G 3:** number of particle measuring more than SMzoo (here 30 pixels)
- **mean grey G: 43:** mean grey level of these "large" particles.

These last data will be utilized to plot the initial profile. The **nb blobs G** represents the number of vignettes (ie objects) to be later identified.

The **nb blobs P-G** data (default setting) is displayed as an analog voltage onto the CTD application when the UVP is connected to a CTD. The number is smoothed and converted in voltage according to the gain (see chapter 8.1.6).

9.2.5.5 BRU and BRU1 files data description

The BRU file contains the measurements and position of each particle counted by the UVP. The BRU files are stored in the process RAW folder and later concatenated and cleaned by Zooprocess during the "Process DAT, BRU and vignette" process.

The UVP5std instruments create BRU files which contains the image date & time while the UVP5hd models provides lighter BRU1 files where the image date & time is absent.

Example of the first two lines: different fields are indicated using different colors corresponding to the different 4 sections of the line

index;image;blob;area;meangrey;xcenter;ycenter;
1;20140625160226_000; 0;84;33;1214; 196;

Detailed description:

- **Index: 1** => the image number, starting at 1 in the RAW dat files.
- **Image: 20140625160226_000** => corresponding image acquisition time (YYYYMMDDHHMMSS_ms) for UVP5std models only
- **Blob: 0 =>** N° of the object in the image
- **Area: 84 =>** Area in pixels
- **Meangrey: 33 =>** mean grey level of the object (8 bits)
- **Xcenter : 1214 =>** X position of its center of gravity
- **ycenter: 196** => Y position of its center of gravity

This BRU file is later simplified and concatenated in a unique file without header. The Image field is removed and the images before the "First Image OK" removed too for standard models. This final "cleaned" file is saved both into the profile folder (work subfolder) and in the Results folder.

9.2.5.6 Header files description

The HDRYYYYMMDDHHMMSS.hdr file downloaded in the raw folder is copied and its extension changed into "txt". It contains a copy of the acquisition mode file utilized to acquire and process the images during the sequence.

See the related chapter: UVP5 acquisition and Process files.

9.2.5.7 Histo file description

The UVP5hd models provides “Histo” synthetic files available in the raw folders. These files contain summarized data from the sequence. These files are utilized to plot the detailed graphs (see above).

9.2.5.8 Additional identification tools

Zooprocess 7.21 and above versions provide direct export tools for the ECOTAXA sorting application.

9.2.6 Piloting UVP5 using a Terminal application

This chapter contains the terminal commands to pilot UVP using a Terminal application (MOTOCROSS). This method is efficient for well-trained users and for trouble shooting.

Be careful when powering the camera. Use any applicable method to avoid switching UVP5 OFF when camera is ON for any purpose. Use stickies or any reminder if you have to move in order to avoid someone else making a mistake.

- RUN sequence using RS232 commands
- Setting time
- Read & Program sequence
- Download data from UVP5
- Delete data in UVP5
- Emergency STOP
- WAKE Up UVP5
- Adjust/test lights (monthly)

The commands are sent to UVP5 by copying them one by one from a text file (copied below). Start application and Turn ON UVP5.

Many others command are located in “UVP5_snXXX_Terminal commands RS232”

9.2.6.1 CAMERA TEST and Data recovery

U013STBY_PRG!
U017START_HD_CAM!
S013WATCHDOG!
=======> Check com with Camera and available space
=======> allow data download and erase in UVP
U016STOP_HD_CAM!
U012RUN_UVP!

9.2.6.2 ON DECK EMERGENCY STOP

=> Connect Sea Cable (green)
=> Start MOTOCROSS.exe
U013STOP_ACQ!
Wait MORE than 1 minute before disconnecting the POWER SHUNT

9.2.6.3 On deck acquisition test / manual sequence

U013PRE_ACQU!
U020STARTACQ_mixtfd!
U013STOP_ACQ!

9.2.6.4 light test

U013STBY_PRG!
U024SET_OUTPUT_00000001!
U024SET_OUTPUT_00000101! => light flashing rapidly (looks continuous)
U024SET_OUTPUT_00000000!

9.2.6.5 Settings general (update config.ini)

U016BAUD_009600!
U024SET_CNA_0_0010_0000!
U020SET_FLAG_1_0030!
U019SET_PPREP_0010!
U012STT_050!
U014SVA_25200!
U018SET_PMIN_0000!
U024SET_PFILT_0200_0001!

9.2.6.6 TIME READING

U013GET_TIME!

9.2.6.7 TIME SETTING

U033SET_TIME_YYYY_MM_DD_HH_MM_SS!

9.2.6.8 To wake up Sleeping system

Enter the following messages without delay between sendings) till you get a STBY_OK return message.

U013STBY_PRG!U013STBY_PRG!U013STBY_PRG!U013STBY_PRG!
U013STBY_PRG!U013STBY_PRG!U013STBY_PRG!U013STBY_PRG!

9.2.6.9 Programming TIME sequence

U013STBY_PRG!
U020SET_FLAG_0_0060!
U018SET_PMIN_0000!
U050SET_SEQ_0000_mixthd_H_MMDDHHMMSS_H_ *MMDDHHMMSS*!
U013READ_SEQ!
U012RUN_UVP!

UVP5 will go into SLEEP mode between casts and wake up automatically about 90 seconds before programmed starting time.

Any configuration file (mixtfd, fullhd, mixthd…) can be utilized in Time mode. Choice will be done according to objectives.

Where MMDDHHMMSS is starting time (time of first image) and *MMDDHHMMSS* is time of last image of the sequence.

9.2.6.10 Programming I/O sequence

U013STBY_PRG!
U020SET_FLAG_0_0060!
U018SET_PMIN_0000!
U050SET_SEQ_0000_fullhd_I_0000000000_I_0000000000!
U013READ_SEQ!
U012RUN_UVP!

Note : this sequence will be run every time you plug the I/O shunt. No need to program it between acquisitions.

9.2.6.11 Programming DEPTH Sequence

U013STBY_PRG!
U020SET_FLAG_1_0030!
U019SET_PPREP_0010!
Choose one of the acquisition method (mixtfd, mixthd, fullhd).
U050SET_SEQ_0000_mixtfd_D_0000000015_D_0000006000! (default)
U050SET_SEQ_0000_mixthd_D_0000000015_D_0000006000! (in case no space on FD)
U050SET_SEQ_0000_fullhd_D_0000000015_D_0000006000! (to record all images on HD)
U013READ_SEQ! (to check)
U012RUN_UVP!

Note : this sequence will be run every time you satisfy the starting criteria (depth). No need to program it again between casts.

9.3 Management of the UVP5 internal memory

The UVP5 have different memory size. This size is limited. It can prevent the start of a sequence (profile) if there is no sufficient space left.

The deletion of the data in the UVP5 may take hours if to filled by many profiles.
We do thus recommend that the data are deleted on a regular base (after processing and backup) instead of trying to keep many profiles in memory.

The remaining capacity of the memory is displayed in Zooprocess every time the camera is started (for download or delete). This information permit to monitor the status of the drives and take mean full decisions in case you encounter difficulties to download data.

The following record issued from a Pilot log indicates the remaining memory of a HD model : 82390 Mo (the space is the same for FD and HD which are two folders of the same drive).

ACTION : DELETE DATA in UVP5

> COM3 OPEN NOW
=> TRYING TO STANDBY UVP5
sent: U013STBY_PRG!
UVP STBY_OK messages received
> Responding time = 1.5 sec.
=> CHECKING CAMERA STATUS
sent: S013WATCHDOG!
sent: S013WATCHDOG!
=> STARTING CAMERA
sent: U017START_HD_CAM!
=> CHECKING if UVP5 CAMERA application is running
ret: U017START_HD_CAM!
=> CHECKING CAMERA STATUS
sent: S013WATCHDOG!
ret: C042WATCHDOG_**82390***0**82390***03*xxxxxxxxxxxx!

FD free space = 82390 Mo (87 %)
HD free space = 82390 Mo (87 %)

CAMERA STARTED & RESPONDING
Remaining casts in FD : 6
Remaining casts in HD : 0

9.4 Typical sequence of operations at sea

The Cruise level and Daily operation are totally under responsibility of the UVP trained operator while the Profile operations can be handled by a trained CTD operator.

9.4.1 At cruise level

The packing of the instrument and the presence of each part will be checked according to the PACKING LIST. It is highly recommended to validate the presence of each item on the provided list (or create a new one if not provided).

The instrument will be assembled on a secured table and its functioning checked (communication, on deck sequence, light alignment, data download, charging…) before installation in the CTD frame.

The installation in the CTD frame is done without the lights to avoid any risk of breakage. The analogue connection is checked with the CTD operator. A complete check is done again (see daily procedure).

The Deck Unit is placed in a dry place, close to the CTD parking area. The Power and Sea cables are set in a convenient way to avoid them being damaged by people working around the CTD.

The CTD operators, ship watchmen and winch men are informed about the UVP deployment procedure and functioning. Their collaboration is essential to avoid functioning mistakes.

The instrument is utilized at sea according to daily and profile procedures.

At the end of the cruise, the instrument is carefully checked. The light unit is removed before the UVP is removed from the CTD frame.

The connectors are carefully checked, cleaned and greased.

A report is written to provide any technical information about the instrument during the cruise.

The datasheets are copied in order to let a PAPER copy in the crate.

The instrument is packed and the PACKING LIST checked again.

9.4.2 Daily

It is not recommended to download data from the instrument after each cast. It would reduce the lifetime of the connectors of the Seacable increasing the risk of failure during the cruise. It is even less useful if the analogue cable is connected to the CTD and allows to display in real time the raw data from the instrument and its correct functioning. The monitoring of the battery voltage after each cast can also confirm the correct functioning of the instrument.

The daily operations permit to:

- Download data from instrument
- Fill In the metadata
- Process data to check quality
- BackUp data
- Check instrument and light beam
- Empty instrument memory
- Charge battery

It is very important that data are processed on a daily base to allow a good QUALITY CHECK of the provided data. This is the ONLY way to be sure that the instrument works correctly and that the data will be usable.

It is also very important to check the lights on a DAILY schedule and note the operation on the datasheets. In case of failure, it will permit to get back to the latest checking date and limit the number of doubtful profiles.

It is also very important to monitor the memory and delete the profiles on a regular base as this operation may take time.

9.4.3 Profile

We present there a typical sequence of operations to be performed for each profile.

Before deployment

- Stop charge and UVP power (if necessary)
- Disconnect the Power cable and connect the Power SHUNT
- Disconnect the Sea cable if connected

CAST/SAMPLE

- Fill in the station datasheet
- Follow protocol according to settings (I/O, Depth…)
    - 22m
    - Back to Surface
    - Start profile
- WARNING :
    - descent to a minimum of 35m if you need to bring the CTD back on deck.
    - redo rinsing if UVP does not start (no signal on CTD display if analogue connected). No need to bring back on deck !

Back on board

- Check that the UVP5 in no longer flashing
    - If NOK (see emergency stop RS232 procedure)
- If OK connect the Power cable in place of the Power SHUNT
- Read and note voltage from Voltmeter on the datasheet => charge if necessary
- Rince instrument
- Visually and manually check the glass tubes of the light unit

10 Additional information and advanced settings

10.1 UVP5 Configuration files and Command list Syntax (S/C/U/V code)

UVP5 contains two computers. One is hosted on its mother board and run all time the UVP5 is powered. It creates or requires three files:

a) Config.ini : mother board settings
b) Sequence.ini : programming of the acquisition
c) Event.log: records all commands received and sent by the UVP. Useful for troubleshooting.

The second is the Camera computer. It starts only in ACTION mode when the camera is powered. It creates or requires two types of files:

a) Profiler.ini : contains the general software configuration settings
b) Acquisition files : defines the image acquisition and process methods

Each RS232 command starts with a letter which assigns an internal device. Below is the meaning of the letters:

U: Message emitted by user in the Terminal application or in Zooprocess. The message is addressed to the mother board controller.

V: Answer from the controller emitted to the user

S: Message emitted by user or by the mother board controller and directed to the camera (camera must be turned on)

C: Answer from the camera

See the UVP5_Messages_RS232.xls file for more detailed information on all commands. The file is located under *UVP5 USB KEY SN00X\Commands Motocross RS232.xls.*

Reminder: the mother board controller is switched on, as soon as the deck unit power button is turned on, or that the power shunt is plugged.

Enter "**app"** to run again UVP5 application or switch in OFF and ON again.

10.2 Event.log visualization

You'll need to run Motocross.exe and then turn ON the UVP external power. Once the UVP is turned on, you'll have 5 seconds to press Q (Quit) to access the controller. (c:/). Then a couple of commands allow you to perform quick control. Here you'll need to type and not just copy and paste.

If you want to record the content of the event.log file to send information for trouble shouting:

- Configure the Terminal application to capture the text that is displayed.
- Enter "**type event.log**" to load the content of the file into your terminal for diagnose or trouble shooting of the UVP5.

10.3 Delete Event.log file

The event.log needs to be emptied once in a while or after 200 profiles. Zooprocess pilot tool will automatically tell you when it is recommended to rename (or delete) the event.log file.

You'll need to run Motocross.exe and then turn ON the UVP external power. Once the UVP is turned on, you'll have 5 seconds to press Q (Quit) to access the controller. (c:/). Then a couple of commands allow you to perform quick control. Here you'll need to type and not just copy and paste.

We recommend renaming the event.log to keep it as a backup for later troubleshooting:

- Enter "**ren event.log YYYYMMDD.old**" where YYYMMDD represent the current day.

After deleting the event.log file, you may encounter difficulties to start sequences if you do not fill a little bit the new event.log that will be automatically created. We thus recommend you to send 2-3 times the following command line after restarting the application typing "**app**".

U013STBY_PRG!U013STBY_PRG!U013STBY_PRG!U013STBY_PRG!U013STBY_PRG!

10.4 Visualizing Mother Board settings

To visualize the settings you'll need to run Motocross (or another Terminal) and then turn ON the UVP. Once the UVP is turned on, you'll have 5 seconds to press Q (Quit) to access the controller. (c:/). Then a couple of commands allow you to perform quick control. Here you'll need to type and not just copy and paste.

Enter "**type config.ini**" to display the parameters and settings recorded in the mother board controller. Note that the corresponding information can be edited in Zooprocess ("EDIT PILOT BOARD CONFIGURATION") and sent to the UVP5 mother board using the Pilot Tools.

Enter "**type sequence.ini**" to display the programming of the UVP5 sequences. Note that the absence of any programming file (sequence.ini) will turn the UVP5 in deep sleeping mode.

10.5 USING the SERIAL UPLINK of SEABIRD 9 CTD

Some SeaBird SBE9 CTDs and corresponding SBE11 deck units are equipped with a serial uplink which allows the transmission of serial messages from the fish to a PC. This serial uplink can be utilized to monitor UVP5 status and data using a terminal application or the Zooprocess monitoring tool (7.20 or above). Depending on the CTD configuration, this uplink is 9600 or 19200 bauds requiring UVP5 baud rate adjustment. The RS232 provided cable for the UVP5 deck unit may not work as the pin 2 and 3 of the SBE11 maybe inverted. We have also observed that a 19200 bauds uplink truncated many of the "V092" or "C110" UVP5 messages.

Hydroptic can provide a specific cable on demand or indicate how to make it yourself.

10.6 Configuration of the analogue output to CTD

10.6.1 Connections

UVP5 can output a voltage on the "ANALOG" connector. This voltage is proportional with the number of particle counted by the camera.

The related PIN OUT of the ANALOG MCBH6M (6 pin male bulkhead) connector is indicated below:

PIN	DESIGNATION/MCBH6M	
1	Analog Output	0-5Vdc
2	GND	0 V
3	CTD AUX POWER	
4	12V TEMP	From UVP5
5	CH1	UVP5 Analog input #1
6	CH2	UVP5 Analog input #2

10.6.2 Checking the analogue output to CTD

The ANALOG output can be tested every time you turn ON UVP5. A increasing ramp of voltage will be applied during 5s, starting 5s after you power the UVP5 ON. This tool is very useful to test the UVP5-CTD interfacing.

This voltage can be:

a) Monitored using a voltmeter
b) Visualized on the CTD display if connected properly.

The output range depends on the GAIN set. 0 to 5V ramp is achieved with the default gain of 20. Lower or faster ramp increase range may be displayed if you change this gain.

10.6.3 Setting gain for the analogue output using Zooprocess

Select the “Set GAIN for ANALOG output” tool in ADVANCED mode and define the maximum particle concentration to be visualized on the analogue output. Higher observed concentration will result in saturating the analogue output to 5 volt with no damage for the UVP5 or the CTD.

The Zooprocess tool provides also an “UVP5_configuration_file.XML” file to be loaded in the SEASAVE application (after version 7.20).

Note that the analogue output should be utilized for the observation of the UVP5 profile and check of the UVP5 status in real time **only**. Zooprocess tools will produce more reliable data (ie UVP data in CTD files are less accurate than what you get using UVP5 recorded files and contains only the total number of recorded particles).

10.6.4 Seasoft xml configuration file for the UVP5

Every time you change the gain in Zooprocess PILOT tool, an updated xml file is created and saved in the “config” folder of the project. This file eases the configuration of the UVP5 data visualization in SEASAVE application.

10.6.5 Calculating gain for the analog ouput to CTD

The output voltage is linked with the number of particles via the following equation:

$$\mathbf{OUT_{(mV)} = 0.25 \times Gain \times N/Vol_{(L)} + Offset}$$

OUT : output voltage in mV (limited to 5 Vdc)
N : number of particle counted in the images (averaged and updated every 10 images. 10 is set by default in the provided mixtfd.ini, mixthd.ini and fullhd.ini files)
Vol : Image volume (from calibration document)
Gain : to be set (Factory setting = 20)
Offset : to be set (Factory setting = 0)

Notes:

a) Gain and Offset must be positive integer! As the output is updated every 10 images the CTD display can show step profiles if the image acquisition frequency drops for any reason.
b) Signal can also be saturated at a maximum of 5Vdc if the Gain is too high for the observed particle concentrations.

Example:
Achieving a 0-5vdc (OUT = 5) output range for a maximum concentration of 1000 particle / L and an image volume of 0.93 L.

$$\mathbf{Gain = Vol \times (5000 - Offset) / (0.25 \times N)}$$

Gain = 0.93 x 5000 / (0.25 x 1000)
Gain = 19

10.6.6 Setting gain for the analog ouput to CTD using Terminal

Gain can be very easily set using Zooprocess Pilot tools in Advanced mode.

Gain can be also be set in Terminal mode sending the following commands:

U013STBY_PRG!
U024SET_CNA_0_GGGG_OOOO!
U012RUN_UVP!
Where:
OOOO = Offset (=0)
GGGG = Gain

10.6.7 Installing UVP5 inside CTD CAROUSEL frame

The UVP5 can be easily installed inside any SEABIRD 32 Carousel frame of 24 bottles of more than 8 Liters. There is no need to remove any bottle as the UVP5 will be placed behind.

An optional set of parts can be ordered for this purpose.

The mounting of the UVP5 in the Carousel is described below.

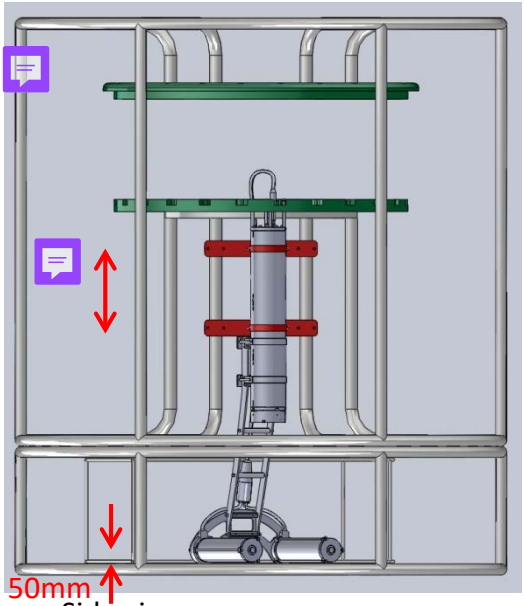
- All stainless steel parts (8) are represented in RED
- All plastic parts (6) are in GREEN

Recommendations:

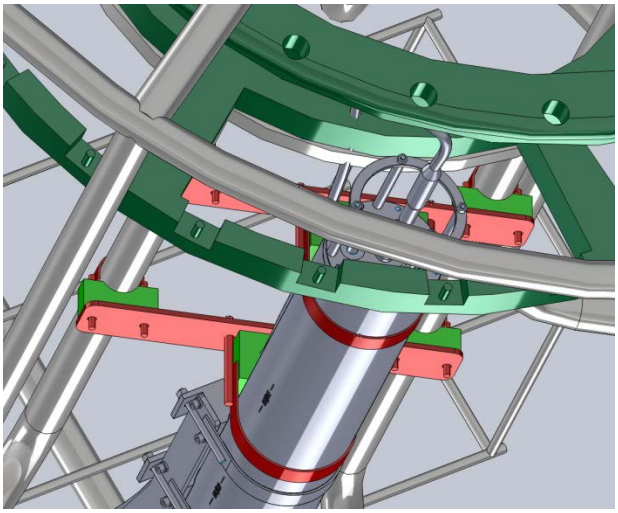
- UVP5 lights will be > 5 cm above lowest part of the Carousel frame to avoid chocks on the ship deck.
- Cable extends must be utilized to ease connexion to the UVP5.
- Care must be applied when installing the UVP5 in carousel. It is recommended to set UVP light system after installing the UVP5 in its berth in order to prevent damaging the light glass tubes.
- UVP5 fixing sets distance should be > 250 mm.

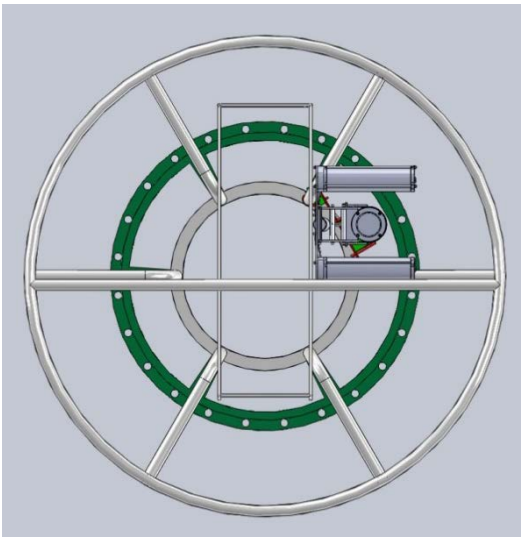
Most of the provided parts can be utilized for other home-made carousels.

10.6.7.1 Mounting the UVP5std in the carousel

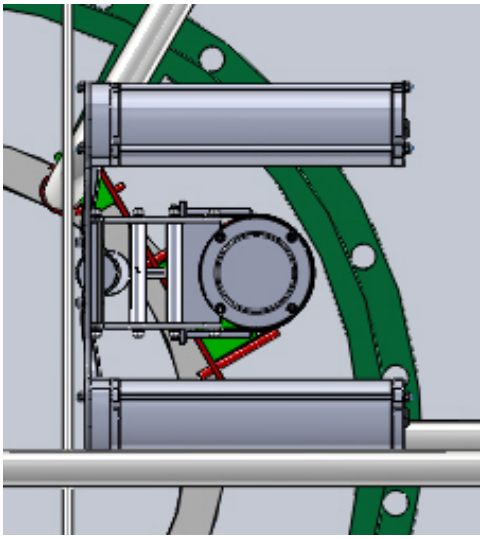


Side view

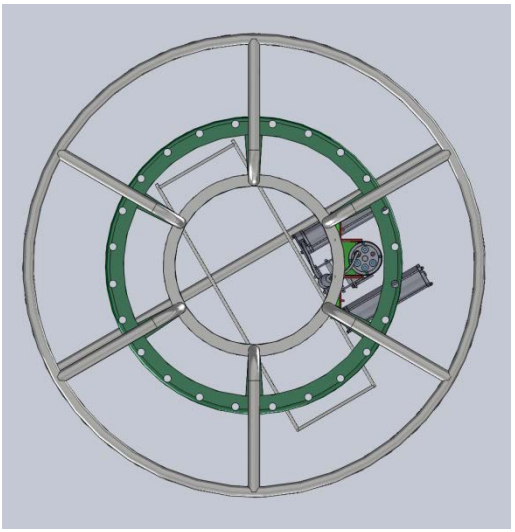

Detail showing all parts



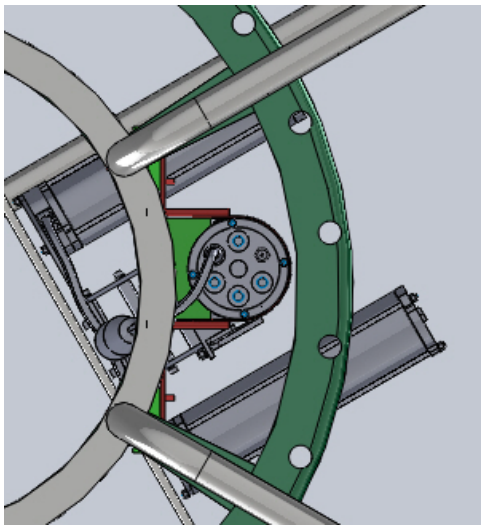
Bottom view: CTD stand orientation



Bottom view detail

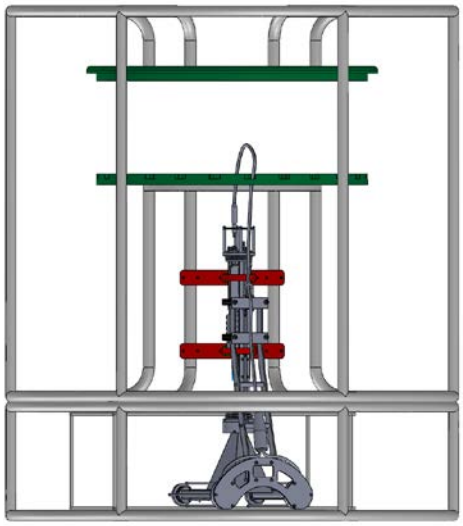


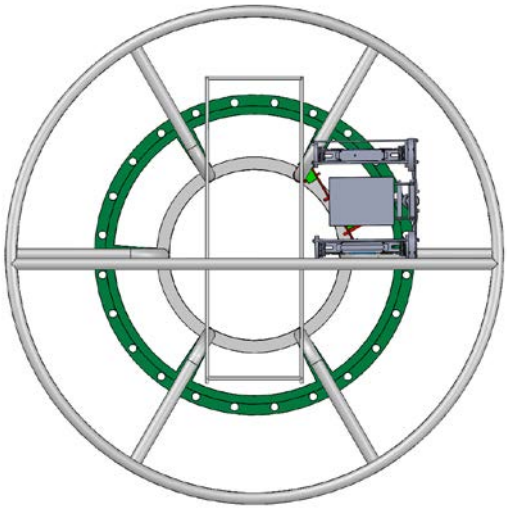
Top view



Top view detail

10.6.7.2 Mounting the UVP5hd in the carousel



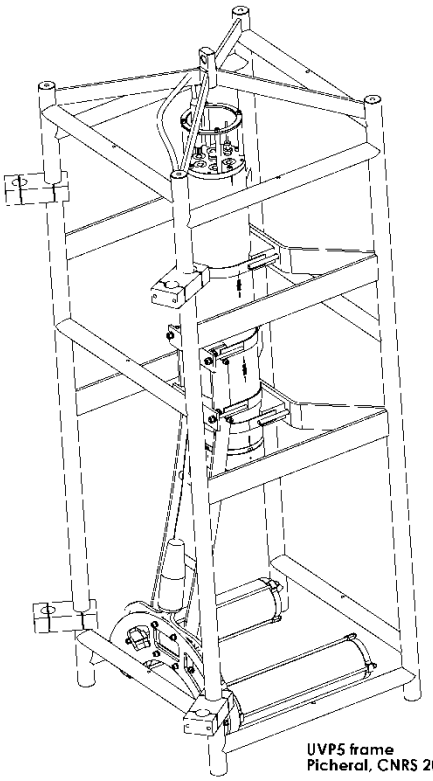


10.6.8 Using UVP5 outside CTD CAROUSEL frame

The UVP5 can be easily installed outside any Carousel frame provided that you have ordered the UVP5 specific frame. This option is convenient for most 12 bottles rosettes which do not have space inside for the UVP5.

There is no need to remove any bottle as the UVP5 will be placed apart.

An optional frame can be ordered for this purpose. This frame can be attached to SEABIRD 12 bottle carousel using the specific nuts (schematic) or to any CTD frame using cords (photo). This frame can also be utilized alone or with a small autonomous CTD attached inside.



UVP5 frame
Picheral, CNRS 2012/12/07



10.7 Safety features

Three main safety features allow the UVP5 to switch to SUSPEND/SLEEP mode (energy saving mode) and thereby interrupt the on-going sequences.

The parameters defining the safety features are monitored and checked at 30 second intervals. If an abnormal value is reached and that the value remains abnormal for two consecutive checks, the SUSPEND mode is activated and a message is sent via the RS232 port.

The UVP5 will then need to be awake by sending many successive STANDBY commands : U013STBY_PRG!U013STBY_PRG!U013STBY_PRG!U013STBY_PRG!U013STBY_PRG!

10.7.1 General Thermal safety feature

A temperature check is performed using the temperature sensor of the mother board. The temperature is not precise but guarantees that no components are operating at temperature levels that could damage them especially when the system is on the deck.

The default maximum temperature value is 50°C since the sensor is not thermally linked to the most fragile component.

10.7.2 Camera Thermal safety feature (UVP5hd only)

The camera of the UVP5hd is cooled to be kept below 22°C to avoid recording image noise. In case the cooling system does not work and the camera temperature raises up to 25°C, the acquisition will stop automatically.

10.7.3 Battery check safety feature

The voltage that power the UVP5 (battery if connected) is checked for the temperature. If the voltage measured reaches levels lower than the recommended value, the system will switch into SUSPEND mode. The curves supplied in the general document for the UVP5 indicate the different voltages and cell charge state.

The recommended default value is 23300 (23.3 Volts) when the system power supply is provided by the internal cells. In the event of the power supply being provided by an external 24V battery this value will need to be modified to adapt to the power supply voltage.

10.7.4 Description of the UVP5 acquisition and process files

The UVP5 camera is provided with defaults acquisition modes defined by the mixtfd.ini, mixthd.ini and fullhd.ini files. These files have been utilized for the calibration of the UVP5 before delivery. You will have to download a copy of these files stored in the UVP5 the first time you run the UVP5 pilot tools of Zooprocess.

Even if we DO NOT recommend to modify these files, they can be edited via the dedicated advanced tool. New ones can also be created from the existing ones. You will have to send these file to the UVP5 camera and you can do it using the advanced dedicated tool too.

Some parameters should not be modified (in RED) if you want to keep the calibration parameters for the size measurements.

10.7.4.1 Downloading the acquisition files using terminal application

Use the procedure "CAMERA TEST and Data recovery" to start the UVP5 camera and access the files in its modfile folder via Windows Explorer and copy them in your project.

10.7.4.2 Downloading the acquisition files using Zooprocess

Use the piloting tools:

- "GET ACQUISITION and INI files from UVP5"

10.7.4.3 UVP5 files for standard models (UVP5std)

[General]
; UVP5std sn0xx : mixtfd
; -1 : error, 0 : Save only, 1 : Process only, 2 : Mixt process, 3 : Full process
TaskType= 2
; -1 : error, 0 : FD, 1 : HD
DiskType= 0
; -1 : error, 0 : BMP, 1 : TIFF, 2 : AVI,3 : JPEG
Filesavetype= 0
; Size of the AVI sequence (only if Filesavetype=2)
AVIFileSize= 0
; Delay between two images (only if Filesavetype=2)
AVISleep= 1000

; cf SONY documentation for SHUTTER
ShutterSpeed= 12
; cf SONY documentation for SHUTTER
ShutterMode= 1
; cf SONY documentation for SHUTTER
ShutterPolarity= 0
; cf SONY documentation for GAIN
Gain= 6
; cf SONY documentation for GAMMA
Gamma= 0
; cf SONY documentation for GAMMA
Threshold= 512

[Sequence]
; 0 : No trigger, 1 : Trigger
TriggerMode= 1

[Processing]
Thresh= 21

SMbase= 0
SMzoo= 30
TimeOut= 1000
BRUoption= 1
EraseBorderBlobs= 0

[Picture]
Choice= 1
Ratio= 3

[TRData]
N= 10
DataMoy= 10110

[Font]
FontPath= C:LOVCAMmodfilegrey.bmp
FontSubstract= 0
SaveAfterSub= 0

10.7.4.4 UVP5 files for High Definition models (UVP5hd) [General]

[General]
; uvp5hd sn2xx : mixtfd

; -1 : error, 0 : Save only, 1 : Process only, 2 : Mixt process, 3 : Full process
TaskType= 2
; -1 : error, 0 : FD, 1 : HD
DiskType= 0
; -1 : error, 0 : BMP, 1 : PNG
Filesavetype= 0

; Exposure time in microS
Exposure= 160 (maybe different for each UVP5hd)
; GainRaw (range : 36-512)
Gain= 200
; Frequence max (1-20 Hz)
Frequence= 20

[Processing]
Thresh= 3
SMbase= 2
SMzoo= 80
TimeOut= 1000
BRUoption= 1
EraseBorderBlobs= 0

[Picture]
; 0 : vignettes, 1 : full image, >1 : vignette or full image if more than N objets

Choice= 0
Ratio= 3

[TRData]
N= 10
DataMoy= 11100

11 Appendix

11.1 Battery certificates (transport regulations)

WILLIAMSON ELECTRONIQUE

NQ185001A

UN 38.3 TEST SUMMARY REPORT

☐ Cell ☒ Battery ☐ Product

Tested type part : 08310Y

Same type part : WILPA2845A

Unique report ID : 7s1p MP176065xlr-12-2018-2

Report date : 2018.04.24

Manufacturer :
Williamson Electronique
6 rue Georges Leclanché, BP 18414
44984 Sainte Luce sur Loire
T. +33 (0) 2.40.18.80.00
https://www.williamson-electronique.fr

Test laboratory :
Saft civil electronics division, Saft SAS
Rue Georges Leclanché, BP 1039
86000 Poitiers
T. +33 (0) 5.49.55.48.48
Litiumsales.fr@saftbatteries.com
https://www.saftbatteries.com

☒ Li-ion battery (rechargeable) ☐ Li-metal battery (primary)

Description : Assembled from seven prismatic MP 176065 xlr component cell with an electronic monitoring circuit as protective device

Battery weight : 1090 g

☒ Nominal Energy : 173.7 Watt-hour **or** ☐ Lithium content : g

☐ Cell ☒ Battery ☐ Product. Model number/part number : 7s1p MP176065 xlr

List of tests conducted	Results (Pass / Fail / N.A.)	Test record reference
T1 : Altitude simulation	PASS	R0038-18
T2 : Thermal test	PASS	R0038-18
T3 : Vibration	PASS	R0038-18
T4 : Shock	PASS	R0038-18
T5 : External short circuit	PASS	R0038-18
T6 : Impacted/crush (cell only test)	PASS	T0141A-17
T7 : Overcharge (N.A for Li-metal only)	PASS	R0038-18
T8 : Forced discharge (cell only test)	PASS	T0141A-17
Battery assembly : ☒ Not applicable. ☐ UN38.3.3 (f) ☐ UN38.3.3 (g)		
Test reference : UN manual of tests and criteria, part III sub-section 38.3. 6th revised edition 2015		

PRODUCT CLASSIFICATION FOR TRANSPORT (According to UN-DGP)

UN Classification / Proper shipping name : **UN3480 / Lithium ion batteries**

Signatory A. Date : 14/12/18
Name : Cécile Burlot
Title : Quality Manager
Signature :

Signatory B. Date : 14/12/18
Name : Thierry Bouessay
Title : Technical director
Signature :

Important! The above signatory / signatories affirm that this document is a true and correct summary of the original individual tests and test data. The original test data is confidential information available to competent state authorities with valid identification and only upon their formal request. Disclosure of the original test data to any other entity upon its request will be considered by Williamson Electronique and, should Williamson Electronique consider this request is with merit, may subject to prior execution of a nondisclosure agreement.

WILLIAMSON ELECTRONIQUE

STATEMENT

April 8th, 2016

Ref: Williamson Electronique – WILPA2093 – 7s1p MP176065

We hereby certify that the power source referenced WILPA2093, assembled by Williamson Electronique from seven Saft MP176065 cell model in serie, configuration known as 7s1p, has a nominal energy of 178.5Wh.

A battery model with the same MP176065 component cell and similar 7s1p electrical configuration has been already constructed by Saft-SBG (P/N: 08310Y) tested by Saft-SBG and compliant with the current applicable "Recommendations on the Transport of Dangerous Goods, Part III, sub-section 38.3, Manual of Tests and Criteria" issued by the United Nations Committee of Experts on the Transportation of Dangerous Goods.

Product Description

Nominal Voltage	26.25 Volts
Nominal Capacity	6.8 Ah
Nominal Energy	178.5 Wh

Product Classification

This lithium battery and/or cell is assigned for transportation to **UN 3480** in accordance with the current **UN Recommendations on the Transport of Dangerous Goods Model Regulations.**

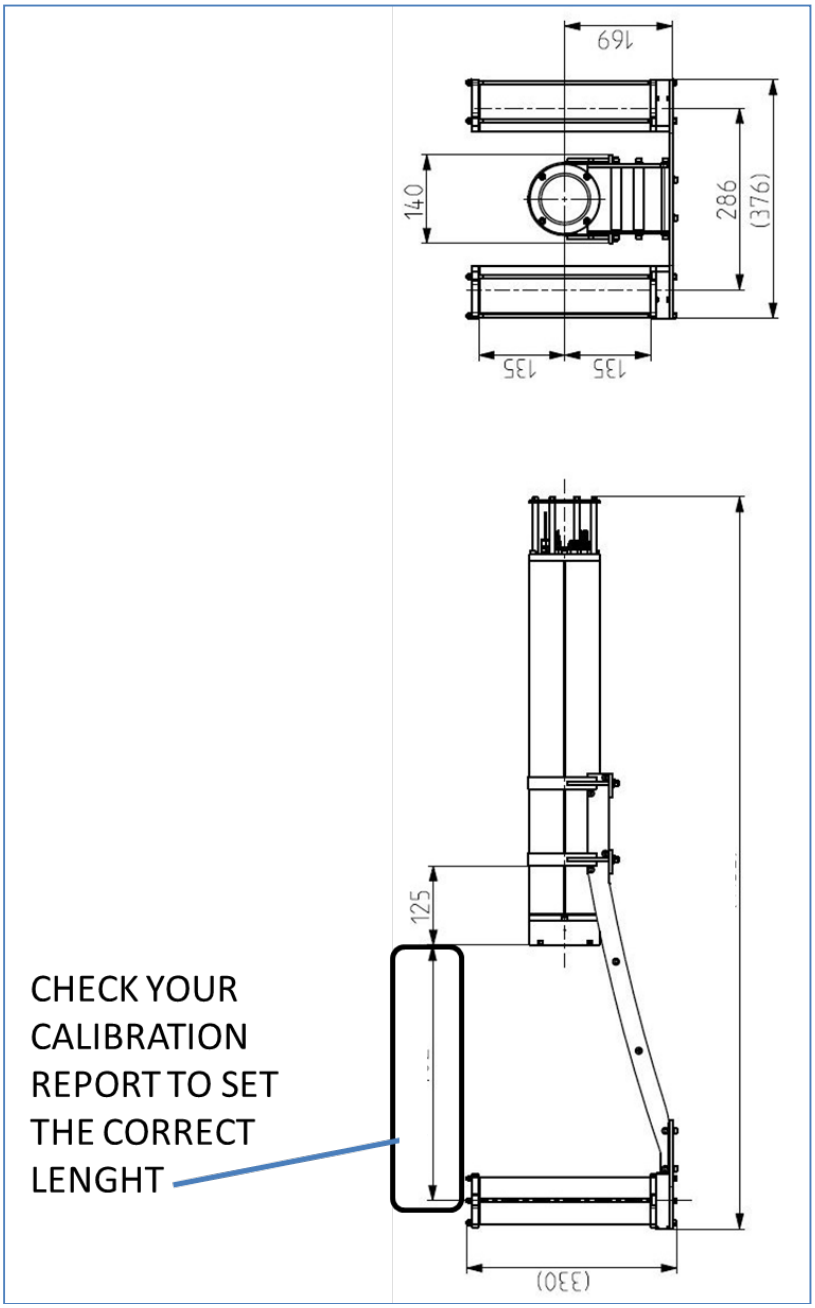
It is your responsibility to comply with the applicable transportation regulations.

Norbert Malin
Director

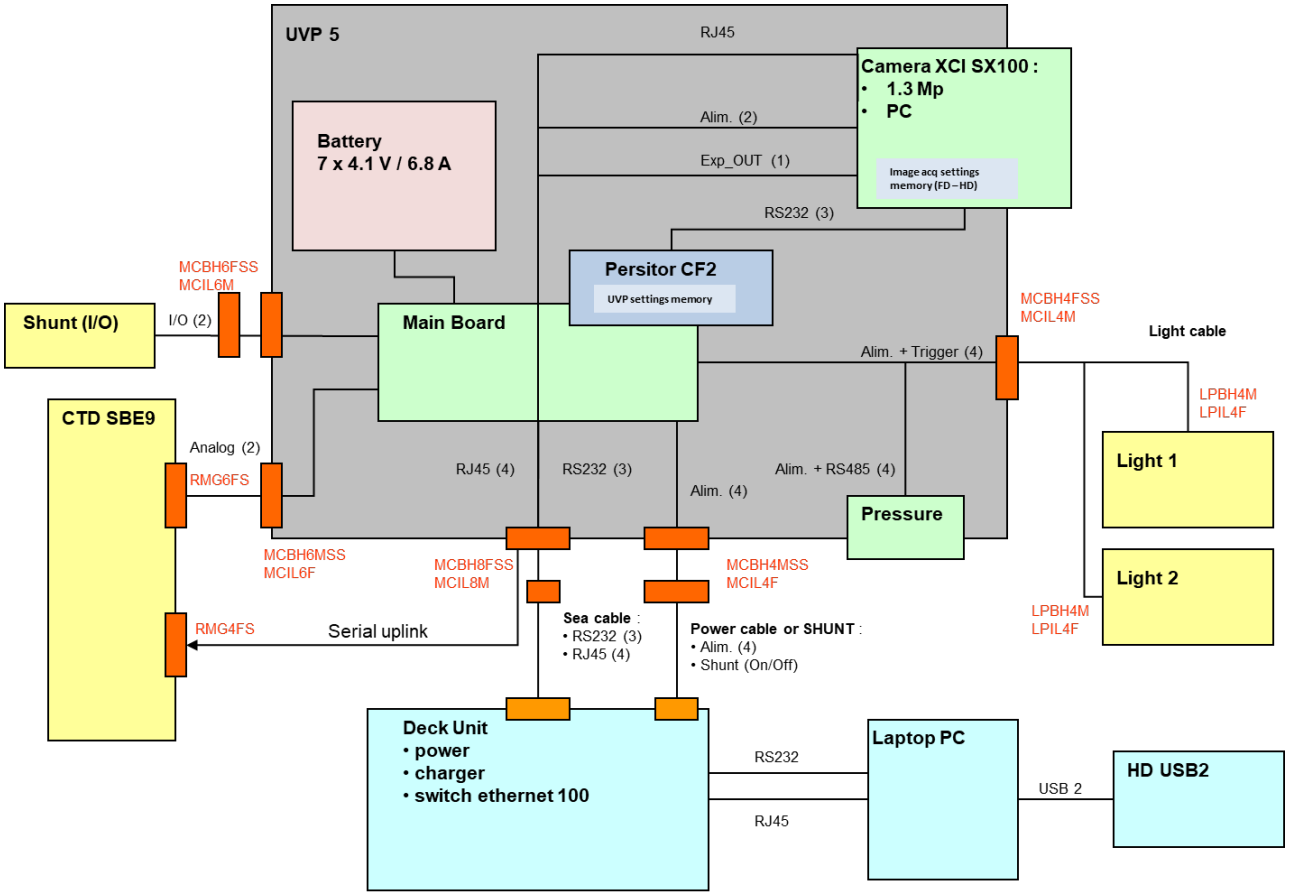
Cécile Burlot
Quality Manager

Williamson Electronique S.A.S - 6/8 rue Georges Leclanché - BP 18414 - 44984 - Sainte-Luce-sur-Loire cedex
Tel : 02 40 18 80 00 - Fax : 02 40 18 80 28
Web : www.williamson-electronique.fr

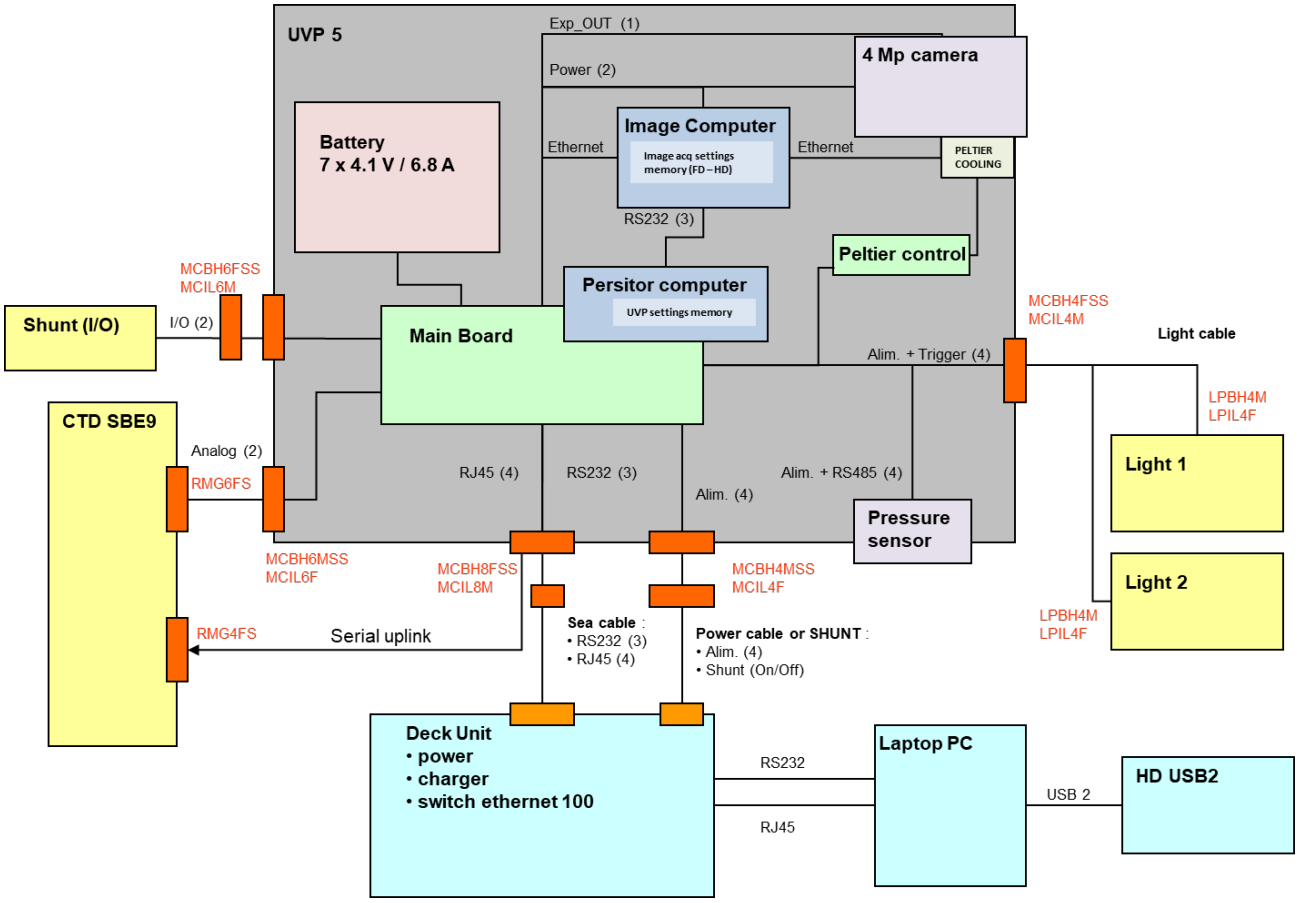
11.2 UVP5std overall dimension



11.3 UVP5std overall diagram



11.4 UVP5hd overall diagram



11.5 UVP5 GENERAL MAINTENANCE

11.5.1 Introduction

This chapter is to give the user basic guideline on how to carry simple maintenance operations on the UVP5 hardware. Rinsing the UVP5 and light beam control are very important and should be performed on regular schedule.

11.5.2 Cleaning the UVP

The UVP must be rinsed with fresh water after each cast.

a) Rinse the UVP entirely
b) Rinse the porthole by inserting water through one of the holes
c) Do not forget to rinse the power shunt and the IO shunt on the side.

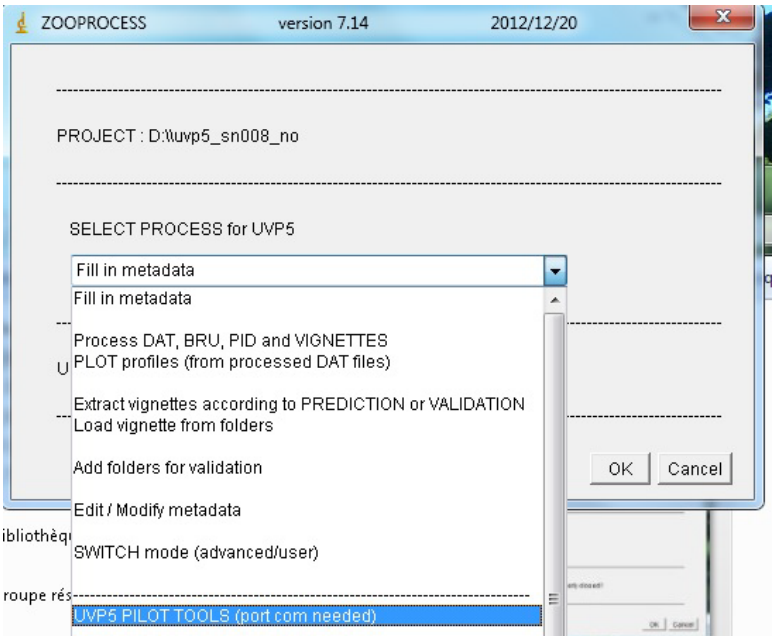


11.5.3 Light beam control

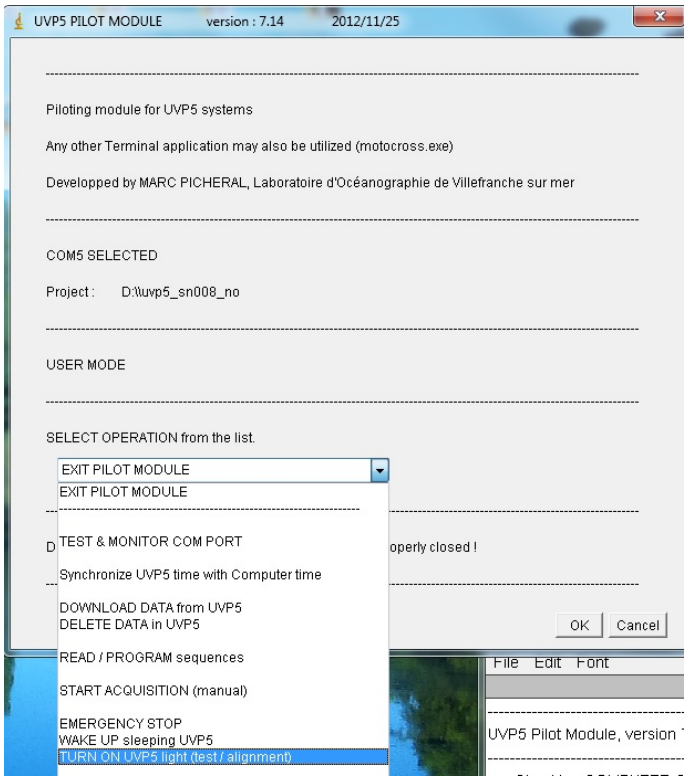
11.5.3.1 Turn lights ON with Zooprocess

Switching the lightings on by using ZOOPROCESS:

1. Open Zooprocess
2. Select UVP5 Pilot Tools



3. Scroll down the pilot module operation list and select <TURN ON UVP5 light (test alignment)>.



Light will be ON during 30s.

11.5.3.2 Checking beam integrity

Even if you get data that look OK, you cannot be sure that abundances are not biased by the failure of some internal parts of a light or by one of the two lights.

This operation will be done **once a day** at sea. It permits to check that both lights work well and that data acquired since you performed the last test are OK.

Turn lights ON as described above and check that the beam issued from each light is continuous (no dead section). Pass your hand between the two lights and check the beam is not interrupted. Restart light if necessary!

11.5.3.3 Controlling the beams parallelism

1. Insert the Tshape jig between both lights and make sure the beams are well positioned
    a. If in water, light beams should be parallel and encompassed between the external 2cm separated lines of the JIG.
    b. If in air, the beams should be focused on the central line.
2. If the beams are not well positioned, carry out the next step to adjust the parallelism

11.5.3.4 Adjusting lights beam parallelism

This operation will be done if you observe a misalignment of the beam or if you need to replace a light.

1. Turn ON light in Zooprocess to detect which light needs to be rotated.

2. Unscrew the 3 nuts from the crescent plate only. Unscrew on a few threads until the glass can rotate.
3. Rotate the lighting glass clockwise or anticlockwise in order to meet the alignment in between the two black lines of the JIG. Tighten the nuts (not too strong) Repeat the operation for the second lighting glass if necessary.





11.5.4 Screws control

Please control the 8 screws from the front socket tap and the black port hole. They shouldn't be loose.





11.5.5 Pressure sensor oil refill

To be performed on a yearly schedule.

Be very careful as the syringe needle MUST NOT damage the sensor membrane at the bottom of the oil container.

Use the syringe and a DC 200 oil bottle from the tool box, follow the steps hereunder:

1) fill the syringe with DC -200 oil from the plastic bottle
2) Screw-lock needle into place

3) Remove pressure capillary nylon fitting and fill cavity with silicon oil (DC200 fluid, with 20 centistokes viscosity)
4) Replace nylon fitting. Oil in cavity should fill capillary as capillary is screwed in
5) The purpose of the nylon fitting and plastic tube is to prevent any salt water from getting into the end cap penetration. The pressure transducer itself is protected from salt water by the oil-filled stainless capillary tube connecting it to the end cap.

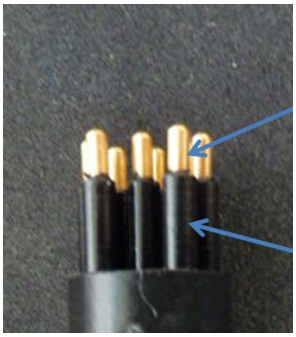



11.5.6 Connectors maintenance

This operation will increase the service life of the connectors and sockets. It is very important to perform it very often on the Power Shunt female connector. High care must be applied on LIGHT and POWER connectors as there is about 30Vdc on a pin which increase the risks of corrosion.

Grease to be utilized: Molykote 44 medium





Female connectors from both the cap and the cables need to be greased from time to time. Drop some grease with your finger in the female holes of the connectors and insert the male plug.

In case a female connector starts to be oxidized, you can use specific connector cleaning products to clean it.

You will clean male connectors using ScotchBrite cleaning pads (NEVER use sand paper!).



11.5.7 UVP5 detailed list of RS232 command

These commands are for the use of highly skilled operators. They permit to utilize the UVP5 for special purposes: moorings, ROVs, AUVs, SUBs and more. Ask Hydroptic!

This detailed list of command is available on demand only.

11.5.8 UVP5 Station datasheet

This example of datasheet is formatted to ease filling metadata in Zooprocess.

Profile DATE (UTC)	
Profile TIME (UTC) *approximate*	
Profile name (HDRYYYYMMDDHHMMSS) *To be read from RAW folder after download*	HDR20
Cruise	
Ship	
Profile Id *Will provide the folder name in work folder (must start with alphanumeric character)*	
Station Id *Name of the station : no associated file naming*	
Bottom depth *For information (missing : 9999)*	
CTD-Rosette reference file *Will ease association of UVP5 data and CTD data*	
Latitude	
Longitude	
First image OK ***To be read in the DAT file of the RAW folder***	
Day / Night	
Wind direction (°)	
Wind speed (Knots)	
Sea state	
Nebulousness (8e)	
Maximum profile depth	
Operator	
Battery voltage (END of CAST) [Volts]	
Light test Y/N (BEFORE cast)	